

ENHANCED REAL-TIME DRONE DETECTION USING YOLO

*A Project Report Submitted in the partial fulfillment of the requirements
for the award of the degree of*

BACHELOR OF TECHNOLOGY IN ELECTRONICS AND COMMUNICATION ENGINEERING

Submitted by

V. KARISHMA

22981A04O4

Under the esteemed guidance of

Dr.V. Keerthi Kiran, M. Tech, Ph.D

Associate Professor

Department of Electronics and Communication Engineering



RAGHU ENGINEERING COLLEGE (Autonomous)

(Affiliated to JNTU Gurajada, Approved by AICTE, Accredited by NBA,
Accredited by NAAC with A+ grade)

Dakamarri, Bheemunipatnam, Visakhapatnam, Andhra Pradesh, India- 531162

www.raghuenggcollege.com

2025-2026

RAGHU ENGINEERING COLLEGE

(Autonomous)

(Affiliated to JNTU Gurajada, Approved by AICTE, Accredited by
NBA, Accredited by NAAC with A+ grade)

Dakamarri, Bheemunipatnam, Visakhapatnam, Andhra Pradesh, India-531162

www.raghuenggcollege.com

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING



CERTIFICATE

This is to certify that the project entitled “**Enhanced real-time drone detection using YOLO**” is the Bonafide record of project work carried out by **V.KARISHMA** (22981A04O4) submitted in the partial fulfillment of the requirements for the award of the degree of **Bachelors of Technology in Electronics and Communications Engineering** during the academic year **2025-2026**.

.

Project Supervisor

DR . V . Keerthi Kiran
Associate Professor
Department of E.C.E
Raghu Engineering College
Visakhapatnam.

Head of the Department

DR . K . Phaninder Vinay
Professor & Head,
Department of E.C.E
Raghu Engineering College
Visakhapatnam.

External Examiner

DISSERTATION APPROVAL SHEET

*This is to certify that the
dissertation titled*
**Enhanced real time drone
detection using YOLO**

By

V. KARISHMA

22981A0404

Is approved for the degree of Bachelor of Technology

.....
Project Supervisor

.....
Internal Examiner

.....
External Examiner

.....
HOD

Date:

DECLARATION

This is to certify that this project titled “**Enhanced real time drone detection using YOLO**” is Bonafide work done by my team, in partial fulfillment of the requirements for the award of the degree Bachelor of Technology and submitted to the Department of Electronics and Communications Engineering, Raghu Engineering College (A), Dakamarri, Visakhapatnam.

We also declare that this project is a result of my own effort and that has not been copied from any one and I have taken only citations from the sources which are mentioned in the references.

This work was not submitted earlier at any other University or Institute for the award of any degree.

Place: VISAKHAPATNAM

Date:

V. KARISHMA (22981A04O4)

ACKNOWLEDGEMENT

We extend our heartfelt gratitude to Raghu Engineering College for providing us with the platform, facilities, and resources to pursue our academic aspirations and successfully accomplish our project work.

We are immensely thankful to **Mr. Raghu Kaldindi**, Chairman of Raghu Engineering College, for his constant encouragement and for providing the necessary institutional and departmental support that enabled us to carry out this project successfully.

Our sincere appreciation goes to **Dr. A. Vijay Kumar**, Principal of Raghu Engineering College, and all the Deans for their valuable guidance and for facilitating the essential infrastructure and academic environment throughout the course of our project.

We express our special thanks to Professor **Dr. K. Phaninder. Vinay**, Head of the Department of Electronics and Communication Engineering, for his continuous support, motivation, and guidance, which played a vital role in the successful completion of this work.

We are deeply grateful to our Final Year Project Coordinator and Supervisor, **Dr. V. Keerthi Kiran**, Associate Professor, Department of Electronics and Communication Engineering, for his insightful suggestions, valuable guidance, and constant encouragement throughout the development of this project.

We would also like to extend our sincere thanks to **Dr. D. Ram Sandeep**, Project Coordinator, for his valuable support and guidance throughout the progress of the project.

Our appreciation also extends to all the faculty members of the Department of Electronics and Communication Engineering for their excellent teaching, encouragement, and guidance, which provided the strong academic foundation necessary for this project.

We would also like to acknowledge the non-teaching staff of the Department of Electronics and Communication Engineering for their cooperation and assistance during the course of this project.

Finally, we express our deepest gratitude to our parents for their unconditional love, constant encouragement, and unwavering belief in us. Their sacrifices and support have been our greatest source of motivation. This work stands as a testament to their guidance and blessings.

Once again, we sincerely thank everyone who directly or indirectly contributed to the successful completion of this project.

V.KARISHMA (22981A04O4)

ABSTRACT

Unmanned Aerial Vehicles (UAVs), commonly known as drones, are increasingly used in various civilian and military applications. However, unauthorized drone activities pose significant security and safety threats, especially in sensitive areas such as airports, defense zones, and public gatherings. Traditional surveillance systems often fail to accurately differentiate drones from birds or other flying objects, leading to false alarms and reduced reliability. This study addresses this challenge by developing an enhanced drone detection system using deep learning-based object detection models.

In this project, we used the YOLOv5 model for detecting and classifying two classes: drone and bird. To improve the accuracy of the system, we combined multiple datasets by merging different train images and their corresponding label files into a single structured dataset. This helped increase the number of training samples and improved the model's ability to generalize better under various conditions. After combining the datasets, we carefully verified the label classes to avoid class mismatch issues and then trained the model using Google Collab with GPU support. Data augmentation techniques such as rotation, scaling, flipping, and brightness adjustment were also applied to further enhance the dataset diversity and increase detection accuracy.

The model was trained and evaluated using standard performance metrics such as precision, recall, and mean Average Precision (mAP). During initial training, some misclassification between drones and birds was observed due to class indexing issues, but after correcting the dataset configuration and retraining the model, the accuracy improved significantly. The final trained model showed reliable performance in distinguishing drones from birds with reduced false detections.

The results demonstrate that combining multiple datasets and properly structuring labels plays a crucial role in improving model accuracy. The enhanced drone detection system developed in this project can be effectively used for real-time surveillance and security monitoring applications. For future work, the system can be extended by adding more diverse environmental images, integrating live video stream detection, and experimenting with advanced YOLO versions for further improvement in speed and precision.

TABLE OF CONTENTS

Certificate	i
Dissertation Approval Sheet	ii
Declaration	iii
Acknowledgement	iv
Abstract	v
Table of Contents	viii - ix
List of Figures	x – xi
List of Abbreviations	xii
List of Tables	xiii
CHAPTER 1: INTRODUCTION	1 - 7
1.1 Background	1 - 3
1.2 Problem Statement	3 - 4
1.3 Objectives	4 - 6
1.4 Scope of the Project	6 - 7
1.5 Applications of Enhanced drone detection	7
CHAPTER 2: LITERATURE REVIEW	8 - 15
2.1 Literature Review	8 - 11
2.2 Existing works	12 - 14
2.3 Summary	14 - 15
CHAPTER 3: SIMULATION SOFTWARE	16 – 21
3.1 Introduction	16
3.2 Hardware Configuration	16
3.3 Software Requirements	17
3.4 Dataset Preparation & Preprocessing	17 - 18
3.5 Model Configuration & Parameters	18 - 19
3.6 Training Workflow	19
3.7 Dataset	19 – 21

3.8 Discussion of Simulation Observations	21
3.9 Summary	21
CHAPTER 4: PROPOSED DESIGN	22 - 36
4.1 Introduction	22
4.2 System Architecture	22 - 26
4.3 Image Annotation	26 - 27
4.4 Dataset splitting	27 - 28
4.5 Image preprocessing	28 - 29
4.6 YOLO models	29 - 30
4.7 Real-Time Object Detection	30 - 32
4.8 Output Visualization	32 - 33
4.9 Performance Evaluation	33 - 35
4.10 Advantages of Proposed System	35 - 36
4.11 Summary	36 - 37
CHAPTER 5: RESULTS AND DISCUSSIONS	38 - 57
5.1 Introduction to Evaluation	38
5.2 Comparison of different YOLO Models	39 - 42
5.3 Training YOLOV5 on Different Datasets	43 - 48
5.4 Enhancement of Cross Datasets	49 - 53
5.5 Comparative Analysis	53 - 55
5.6 Summary of Results and Discussions	55 - 57
CHAPTER 6: CONCLUSION	58 - 60
CHAPTER 7: FUTURE SCOPE	61 - 62
REFERENCES	63 - 66

List of Figures

	Page no
Figure 1.1 Drone Detection	3
Figure 3.1 YOLOV5 Architecture	17
Figure 3.7.1 Sample Drone Dataset Images	20
Figure 3.7.2 Sample Drone Images	21
Figure 4.2 System Architecture	23
Figure 4.6 YOLOV5 Model Architecture	30
Figure 4.7 Real time Object detection	30
Figure 5.2.1 Performance metrics(precision, recall, mAp@0.5 and mAP@0.5:0.9) for YoloV5	38
Figure 5.2.2 Performance metrics(precision, recall, mAp@0.5 and mAP@0.5:0.9) for YoloV8	39
Figure 5.2.3 Performance metrics(precision, recall, mAp@0.5 and mAP@0.5:0.9) for YoloV10	40
Figure 5.2.4 Performance metrics(precision, recall, mAp@0.5 and mAP@0.5:0.9) for YoloV11	40
Figure 5.2 Detected Images of Different yolo models	42
Figure 5.3.1 Performance metrics(precision, recall, mAp@0.5 and mAP@0.5:0.9) of Drone detection 1	43
Figure 5.3.2 Performance metrics(precision, recall, mAp@0.5 and mAP@0.5:0.9) of Drone detection 2	44
Figure 5.3.3 Performance metrics(precision, recall, mAp@0.5 and mAP@0.5:0.9) of Drone detection 3	44
Figure 5.3.4 Performance metrics(precision, recall, mAp@0.5 and mAP@0.5:0.9) of Drone detection 4	45
Figure 5.4 Enhanced value graphs	49
Figure 5.4.1 Detected Drone images of Enhanced Drone – Dataset1	50

Figure 5.4.2	Detected Drone images of Enhanced Drone- Dataset2	51
Figure 5.5.1	Precision comparison before and after applying dataset enhancement techniques, showing improved detection accuracy across datasets.	55
Figure 5.5.2	Recall comparison before and after dataset enhancement, indicating improved object detection sensitivity.	55
Figure 5.5.3	mAP@0.5 comparison before and after dataset enhancement, reflecting improved detection performance at IoU threshold 0.5	55
Figure 5.5.4	mAP@0.5:0.95 comparison before and after dataset enhancement, demonstrating improved model generalization across multiple IoU thresholds	55

List of Abbreviations

YOLO	You Only Look Once
mAP	Mean Average Precision
mAP@0.5	Mean Average Precision at IOU threshold 0.5
mAP@0.5:0.9	Mean Average Precision across IOU thresholds 0.5 to 0.9
IOU	Intersection over Union
F1 Score	Harmonic Mean of Precision and Recall
PR Curve	Precision-Recall Curve
P-Curve	Precision Curve
R-Curve	Recall Curve
RGF	Rolling Guidance Filter
MATLAB	Matrix Laboratory

List of Tables

	Page no
Table 2.2.1: Difference between Existing work and proposed methodology	13
Table 5.2.1: Comparison of different models by using Drone dataset1	38
Table 5.3.1: Trained values of multiple drone datasets using YOLOV5 Model	42
Table 5.5.1: Enhanced dataset values	47
Table 5.6.1: Comparison of original and enhanced datasets	55

CHAPTER 1

INTRODUCTION

1.1 Background

In recent years, drones have become a common part of everyday life. What was once considered advanced technology is now easily available and widely used across many fields. From capturing aerial photos and videos to helping farmers monitor crops, drones have made many tasks quicker and more efficient. They are also used in surveillance, parcel delivery, disaster management, and traffic monitoring. Their ability to fly over large areas, access difficult locations, and provide real-time data makes them highly useful in both commercial and public sectors.

However, along with these advantages, the increasing use of drones has also raised serious security concerns. While many drones are used for legitimate purposes, there are situations where they are operated without permission or for harmful activities. Unauthorized drones entering sensitive areas such as airports, military zones, or crowded public places can create dangerous situations[1]. They may interfere with aircraft operations, cause accidents, or be used for illegal surveillance. Because of these risks, it has become important to develop systems that can continuously monitor airspace and detect drones in a reliable way.

Detecting drones in real-world conditions is not as simple as it sounds. The outdoor environment is highly dynamic, and objects in the sky can appear differently depending on distance, lighting, and background. In many cases, drones are not clearly visible and may appear very small. This makes it difficult for detection systems to capture enough details for accurate identification. In addition, drones often look similar to birds, especially when viewed from a distance or under unclear weather conditions. Their size, shape, and movement patterns can sometimes be confusing, which increases the chances of incorrect detection.

Traditional detection methods such as radar systems, acoustic sensors, or basic image processing techniques have been used in the past. While these methods can detect the presence of flying objects, they are not always capable of accurately identifying them. As a result, they often generate false alarms, especially in environments where birds are present. This reduces the effectiveness of such systems in practical applications.

With the rapid growth of Artificial Intelligence and deep learning, more advanced and reliable solutions have become possible[2]. Computer vision techniques allow machines to analyze images and videos in a smarter way and identify objects with better accuracy. Modern object detection models can process data in real time, making them suitable for surveillance tasks where quick decisions are required. Among these models, YOLO-based approaches are widely used because of their speed and efficiency. In our work, we explored different versions such as YOLOv5, YOLOv8, YOLOv10, and YOLOv11, and found that YOLOv5 provides more consistent and reliable performance for our application.

In this project, we present an Enhanced Drone Detection System that uses deep learning-based object detection techniques. The goal is not only to detect drones but also to accurately distinguish them from birds under different environmental conditions. To achieve this, we used a combination of datasets and improved their quality through proper labeling and preprocessing. This helped the model learn better and perform more effectively in real-world scenarios.

The proposed system focuses on improving detection accuracy, reducing false positives, and maintaining stable performance in real-time conditions. It is designed to handle variations in lighting, background complexity, and object distance, making it more practical for actual deployment. Compared to traditional methods, this approach is more flexible and can be easily adapted to different environments.

Another key focus of this work is to make the system simple to implement while still maintaining strong performance. Instead of depending on complex hardware or multiple sensors, the approach mainly relies on visual data, which makes it more cost-effective and easier to deploy. The model is trained in such a way that it can adapt to different scenarios without requiring frequent retraining. Care was also taken to ensure that the system remains lightweight so that it can run on standard computing devices without high-end requirements[3]. This makes the proposed solution more practical for real-time use, especially in applications where resources are limited but continuous monitoring is needed.

Overall, this project aims to develop a reliable and efficient drone detection system that can be used in real-world applications such as airport security, border monitoring, and public safety. With further improvements, the system can be extended to support live video

surveillance and automated alert systems, making it a valuable tool for modern security needs.

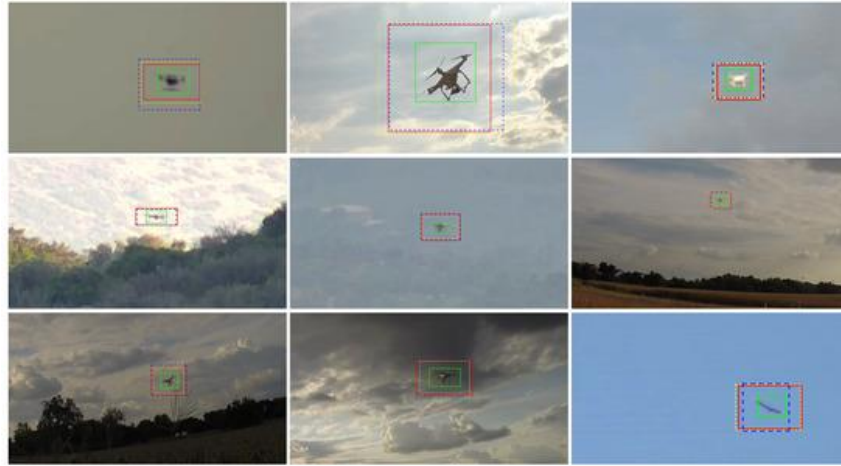


Figure 1.1: Drone detection

1.2 Problem Statement

Even though many object detection systems are available today, detecting drones accurately in real-world situations is still not an easy task. One of the biggest difficulties is telling the difference between drones and birds. From a distance, both can look almost the same in size and shape, and sometimes even their movement patterns are quite similar. Because of this, many systems get confused and end up giving wrong predictions.

Another issue is that drones usually appear very small in the sky, especially when they are far from the camera. When the object is too small, it becomes harder for deep learning models to extract clear features, which directly affects detection accuracy. On top of that, the background in outdoor environments is often complex. Things like clouds, trees, buildings, and other moving objects create noise in the image, making it even more difficult for the model to focus only on the drone.

Environmental conditions also play a big role[4]. Changes in lighting, such as very bright sunlight or low-light conditions, can reduce image clarity. Weather conditions like fog, rain, or dust further make detection unreliable. Sometimes, motion blur caused by fast movement of drones or camera shake also affects how clearly the object is captured. All these factors together reduce the overall performance of detection systems in real-world

scenarios.

Traditional methods like radar or simple sensor-based systems are not intelligent enough to handle these challenges. They often detect any flying object as a potential threat, which leads to a large number of false alarms. This makes such systems less practical, especially in areas where birds are common.

In our project, we also faced several challenges during the initial training phase. The model was frequently misclassifying birds as drones and drones as birds. After analyzing the problem, we found that this was mainly due to limited dataset size and imbalance between the number of drone and bird images. While merging multiple datasets, there were also issues with incorrect label indexing and inconsistent annotation formats. These small mistakes had a big impact on the model's learning process.

Another important observation was that the dataset did not have enough variety in terms of backgrounds, angles, and environmental conditions. Because of this, the model was not able to generalize well when tested on new images[5]. It performed well on training data but struggled in real-world scenarios. This clearly showed the importance of proper dataset preparation and careful handling of labels.

These challenges highlighted the need for a more reliable and improved system. Therefore, this project focuses on developing an enhanced deep learning-based drone detection system that can reduce false positives, improve classification accuracy, and handle small object detection more effectively. The aim is to build a system that works consistently in different environmental conditions and can be used for real-time surveillance applications without frequent errors.

1.3 Objectives

The main objective of this project is to develop an enhanced drone detection system using deep learning techniques that can accurately identify drones and clearly distinguish them from birds. In real-world surveillance, one of the common problems is misclassification, where birds are detected as drones or actual drones are missed. So, a key focus of this work is to reduce such errors and minimize false alarms. At the same time, the system is expected to provide fast and reliable performance so that it can be used in real-time monitoring situations.

During the development of the project, we explored and tested multiple versions of YOLO, including YOLOv5, YOLOv8, YOLOv10, and YOLOv11. Each of these models has its own advantages, so we trained them using the same dataset to compare their performance fairly. After evaluating the results, we found that YOLOv5 performed more consistently for our specific task. It provided a better balance between accuracy and speed and was more effective in distinguishing drones from birds. Based on these observations, YOLOv5 was selected as the final model for our system.

Important objective is to improve the overall detection accuracy by using a diverse and well-prepared dataset. Instead of relying on a single dataset, we combined multiple drone and bird image datasets from different sources. This helped in increasing the variety of images in terms of background, lighting, and object appearance. While merging the datasets, we carefully checked all label files to ensure correct class indexing and proper annotation format. Fixing these issues was necessary because even small labeling errors can affect the model's learning process. In addition, we applied data augmentation techniques such as flipping, rotation, and brightness adjustments to further improve the model's ability to handle different real-world conditions.

The project also aims to develop a system that can generalize well beyond the training data. In simple terms, the model should not just perform well on known images but also give accurate results when tested on new and unseen data[6]. This is important because real-world environments are always changing. Factors such as weather conditions, lighting variations, object distance, and background complexity can affect detection. To handle this, we focused on improving the model's adaptability and reducing overfitting during training.

Another objective is to ensure that the system is efficient enough for real-time applications. The model should be able to process images quickly without compromising accuracy. This is especially important in surveillance scenarios where continuous monitoring is required and decisions need to be made instantly.

Finally, the performance of the system is evaluated using standard metrics such as precision, recall, and mean Average Precision (mAP). These metrics give a clear understanding of how well the model is detecting drones and how effectively it is avoiding false detections. By analyzing these values, we can ensure that the system meets the expected performance levels.

The overall goal of this project is to develop a robust, scalable, and practical drone detection system that can be used in real-world applications. In the future, the system can be extended to support real-time video surveillance and can be integrated into security systems in airports, border areas, and public safety environments[7]. This makes the project not only technically useful but also highly relevant for modern security needs.

1.4 Scope of the project

The scope of this project mainly focuses on building an enhanced drone detection system that can accurately identify drones and differentiate them from birds using deep learning techniques. To achieve this, we collected drone and bird image datasets from different sources and combined them into a single, well-organized dataset. While doing this, special care was taken to ensure that all images and their corresponding label files were correctly matched. Issues such as incorrect class indexing and inconsistent annotation formats were identified and fixed, as these small errors can significantly affect the model's learning process. This step played an important role in improving detection accuracy and reducing the misclassification problems that were observed during the early stages of training.

The implementation part of the project involves several important steps. Before training the model, the dataset was preprocessed to make it suitable for learning. We applied different data augmentation techniques such as rotation, flipping, scaling, and brightness adjustments. These techniques help in increasing the variety of training data and make the model more capable of handling different real-world conditions[8]. The object detection model was then trained using the YOLOv5 architecture along with pretrained weights, which helped in faster convergence and better performance. During training, we carefully monitored the model's progress by analyzing loss values and performance graphs to ensure that the model was learning correctly and not overfitting.

Another part of the scope includes properly configuring the dataset settings, such as preparing the YAML file, which defines the dataset structure and class details required for training. After training, the model was evaluated using standard performance metrics like precision, recall, and mean Average Precision (mAP). These metrics helped us understand

how well the model is detecting drones and how effectively it is avoiding false detections. The current scope of the project is limited to detecting two classes, namely drones and birds, using image-based detection under different environmental conditions[9]. However, the system is designed in such a way that it can be extended in the future. For example, it can be integrated with real-time video surveillance systems to monitor live camera feeds in areas such as airports, defense zones, and other restricted locations. There is also scope for further improvements, such as enhancing the detection of very small or distant objects, improving performance under extreme weather conditions, and deploying the model on embedded or edge devices for faster and more efficient real-time operation. These extensions can make the system more practical and useful in real-world applications.

1.5 Applications of Enhanced Drone Detection

Airport Security : The system can be used near airports to detect unauthorized drones entering restricted airspace and prevent accidents with aircraft.

Military and Defense Surveillance : It helps in monitoring borders and sensitive military areas by detecting enemy or suspicious drones in real time.

Public Safety Monitoring : The model can be used in crowded places such as stadiums, public events, and government buildings to identify drones that may pose security threats.

Smart City Surveillance Systems : The drone detection model can be integrated with smart city monitoring systems to track and control drone activities in urban areas.

Border Security : Governments can use this system to detect drones used for smuggling or illegal surveillance across borders.

Event Security Management : During large public events or VIP gatherings, the system can identify unauthorized drones to ensure safety.

CHAPTER 2

LITERATURE REVIEW

2.1 LITERATURE REVIEW

J. Redmon et al.[10] have focused on introducing the YOLO algorithm for real-time object detection. The authors proposed a unified detection framework that processes the entire image using a single neural network and predicts bounding boxes and class probabilities simultaneously. The study highlights the advantage of YOLO in terms of high detection speed and real-time performance compared to traditional object detection methods. The approach significantly reduces computational complexity and makes it suitable for applications such as surveillance, autonomous systems, and aerial object detection including drones.

J. Redmon and A. Farhadi et al.[11] have focuses on improving the YOLO object detection framework by introducing YOLO9000, which combines object detection and classification datasets to detect more than 9000 object categories. The authors proposed a joint training mechanism that allows the model to learn from multiple datasets simultaneously. The study improves both accuracy and detection capability while maintaining high processing speed, making it suitable for large-scale object detection applications .

A. Bochkovskiy et al.[12] have focuses on improving the performance of object detection systems by introducing the YOLOv4 architecture. The authors implemented several optimization techniques such as data augmentation, feature pyramid networks, and improved training strategies to increase detection accuracy. The study demonstrates that YOLOv4 achieves a good balance between detection speed and accuracy, which makes it suitable for real-time applications including aerial object detection and surveillance systems.

G. Jocher et al. [13] have focuses on developing an efficient and easy-to-use object detection framework using the YOLOv5 architecture. The model provides improved training performance, faster inference speed, and flexible deployment capabilities. YOLOv5 supports multiple model sizes and provides tools for dataset training, evaluation, and deployment. The framework has been widely used for various computer vision

applications such as traffic monitoring, security surveillance, and drone detection systems.

A. Rozantsev et al. [14] have focuses on detecting small flying objects such as drones using computer vision techniques. The authors proposed a method that analyzes motion patterns and appearance features of flying objects captured by a moving camera. The system improves detection accuracy by combining motion information with object recognition techniques. The study demonstrates that computer vision approaches can effectively detect aerial objects even in complex backgrounds.

S. K. Shandilya et al.[15] have focuses on developing a dataset specifically designed for distinguishing drones from birds in aerial images. The dataset contains annotated images of drones and birds captured in different environments. The main objective of the study is to reduce false detections in drone monitoring systems. By providing labeled training data, the dataset helps deep learning models learn the visual differences between drones and birds, improving detection accuracy.

H. Rakshit et al.[16] have focuses on using deep convolutional neural networks to detect drones in aerial surveillance applications. The proposed model extracts visual features from images and classifies objects based on their structural characteristics. The study highlights the challenges involved in detecting small drones due to their size and similarity to birds. The results demonstrate that deep learning methods can significantly improve drone detection performance compared to traditional techniques.

M. K. Kabir et al.[17] have focuses on detecting drones in real-time video streams using deep learning and image processing techniques. The authors used the YOLOv7 model to identify drones from surveillance footage captured in various environments. The system was trained on drone datasets and tested on real-time videos to evaluate its performance. The results show that the proposed system achieves high precision and recall while maintaining fast detection speed.

D. Kaur et al.[18] have focuses on improving drone detection systems by developing a specialized dataset for distinguishing drones from birds. The authors introduced a YOLO-based detection model trained on the BirDrone dataset. The study emphasizes the importance of dataset quality and class balance in improving classification accuracy. The results demonstrate improved detection performance in aerial surveillance scenarios.

R. Laroca et al.[19] have focuses on improving the detection of small drones using

multi-scale image processing techniques and data augmentation strategies. The authors enhanced the training dataset by applying transformations such as scaling, rotation, and brightness adjustment. These techniques help the model detect drones under different environmental conditions. The study shows that multi-scale feature extraction improves detection accuracy for small aerial objects.

W. Liu et al.[20] have focuses on developing a fast object detection framework known as the Single Shot MultiBox Detector (SSD). The authors proposed a deep learning model that detects objects in images using a single neural network without requiring region proposal stages. The system predicts bounding boxes and class probabilities simultaneously, which significantly improves detection speed. The model is capable of detecting objects at multiple scales and has been widely used for real-time applications such as video surveillance and aerial object detection.

S. Ren et al .[21] have focuses on improving object detection accuracy using Region Proposal Networks (RPN). The authors introduced the Faster R-CNN architecture, which integrates region proposal generation and object detection into a single network. The model improves detection accuracy while reducing computation time compared to earlier methods. The study demonstrates that deep learning-based object detection frameworks can effectively detect objects in complex environments.

J. Deng et al.[22] have focuses on creating a large-scale image dataset known as ImageNet for training and evaluating computer vision models. The dataset contains millions of labeled images organized into thousands of object categories. ImageNet has become one of the most important datasets in deep learning research and has been widely used for training convolutional neural networks. The availability of such large datasets has significantly improved the performance of object detection and classification models.

K. Simonyan et al.[23] have focuses on improving image recognition accuracy by developing very deep convolutional neural network architectures. The proposed VGG network uses multiple convolutional layers to extract detailed image features. The study demonstrates that deeper networks can improve classification performance and feature extraction capability. These techniques have influenced many modern object detection models used in computer vision applications.

C. Szegedy et al.[24] have focuses on improving deep learning models for image

recognition by introducing the Inception architecture. The proposed model uses parallel convolution layers with different filter sizes to capture features at multiple scales. This design improves both computational efficiency and detection accuracy. The Inception model has contributed to the development of many advanced deep learning architectures used for object detection.

T. Y. Lin et al. [25] have focuses on developing a large dataset known as the COCO dataset for object detection, segmentation, and image captioning tasks. The dataset contains thousands of images with detailed annotations for different object categories. It helps researchers train and evaluate object detection models in real-world scenarios. The dataset has been widely used in developing modern deep learning detection algorithms.

F. Chollet et al. [26] have focuses on explaining deep learning techniques and neural network architectures used in modern artificial intelligence applications. The book discusses concepts such as convolutional neural networks, image classification, and object detection. It provides practical insights into building deep learning models using Python frameworks. These concepts are widely applied in computer vision systems including drone detection.

I. Goodfellow et al. [27] have focuses on the theoretical and practical aspects of deep learning techniques. The book explains neural networks, optimization algorithms, and training methods used in modern machine learning models. It provides a strong foundation for understanding deep learning applications such as image recognition, object detection, and computer vision systems.

R. Szeliski et al. [28] have focuses on the fundamental concepts of computer vision and image processing techniques. The book discusses algorithms for object detection, feature extraction, and image analysis. These concepts form the foundation of many modern computer vision systems used in surveillance and object detection applications.

D. G. Lowe et al. [29] have focuses on developing the SIFT (Scale Invariant Feature Transform) algorithm for detecting and describing image features. The technique extracts distinctive key points that remain stable under changes in scale and orientation. These features are useful in object recognition and image matching tasks. The research has influenced many computer vision methods used in object detection systems.

2.2 Existing System

- Existing drone detection systems primarily rely on traditional surveillance methods such as radar systems, acoustic sensors, and basic computer vision techniques to identify flying objects in the airspace.
- Some systems also use conventional image processing methods like motion detection, background subtraction, and basic feature extraction to detect drones in images or video streams.
- While these methods are simple and easy to implement, they suffer from several limitations:
 - Poor ability to detect small drones at long distances
 - Difficulty in distinguishing drones from birds or other flying objects
 - Reduced detection accuracy in complex backgrounds such as clouds, trees, or buildings
 - Sensitivity to environmental conditions such as lighting changes, fog, and weather variations
 - High false detection rates in real-world surveillance scenarios

Many existing models also struggle to accurately classify drones due to limited datasets and insufficient training data[30]. In several cases, drones are incorrectly identified as birds because of their similar appearance and movement patterns, which reduces the reliability of drone monitoring systems.

2.2.1 Proposed System

- The proposed system uses the YOLOv5 deep learning model to detect drones in images and video streams.
- The model is trained using a combined dataset of drone and bird images to improve classification accuracy.
- YOLOv5 processes the entire image in a single step, enabling fast and real-time object detection.
- The system can accurately distinguish drones from birds and other flying objects, reducing false detections.

- It performs better in different environmental conditions such as lighting variations and complex backgrounds.

The proposed approach improves the accuracy, speed, and reliability of drone detection compared to traditional detection methods.

Key Differences Between Existing and Proposed Systems

Table 2.2.1 Difference between existing work and proposed system:

Aspect	Existing System	Proposed System (YOLOv5 Based)
Automation Level	Semi-automated systems using traditional image processing and limited machine learning techniques.	Fully automated deep learning-based detection using the YOLOv5 model.
Accuracy	Moderate accuracy and often produces false detections when drones appear similar to birds or small objects.	High detection accuracy due to training with mixed drone and bird datasets using YOLOv5.
Detection Speed	Slower detection speed because some systems process images in multiple stages.	Faster real-time detection since YOLOv5 processes the entire image in a single pass.
Adaptability	Performs poorly in complex environments such as cloudy weather, long distance, or low lighting.	More robust to environmental variations like lighting changes and background complexity.
Object Differentiation	Difficulty distinguishing between drones and birds, leading to misclassification.	Improved classification because the model is trained with both drone and bird datasets.
Dataset Usage	Uses limited or single datasets, reducing model performance.	Uses combined datasets which improves training quality and detection capability.
Scalability	Limited scalability and often used only in research	Easily scalable and can be trained with larger datasets

	prototypes.	for improved performance.
Implementation Platform	Many systems require expensive hardware or specialized equipment.	Implemented using Python, PyTorch, and Google Colab, making it cost-effective.
Detection Capability	Limited ability to detect small drones or drones in complex backgrounds.	YOLOv5 can detect small drones more effectively using deep learning features.
Application	Mostly experimental research systems with limited practical use.	Suitable for real-time applications such as airport security, surveillance, and restricted area monitoring.

2.3 Summary

From the study of different research papers, it can be understood that drone detection has become an important research area due to the increasing use of unmanned aerial vehicles in many fields[31]. Earlier systems mainly used traditional image processing techniques such as motion detection, background subtraction, and basic feature extraction. These methods were easy to implement but often produced inaccurate results, especially when the drone was small or when the background was complex. With the development of deep learning technologies, researchers started using object detection algorithms such as YOLO, Faster R-CNN, and SSD for detecting aerial objects. Among these methods, YOLO-based models are widely used because they provide faster detection and better performance in real-time applications. Many studies have shown that deep learning models can improve the accuracy of drone detection compared to traditional approaches. However, some challenges still exist, such as distinguishing drones from birds and detecting drones in different environmental conditions[32]. To overcome these issues, researchers have started using larger and more diverse datasets for training detection models. Based on the analysis of these studies, deep learning-based object detection models

provide an effective solution for drone detection tasks. Therefore, the proposed work focuses on using the YOLOv5 model trained on a combined dataset to improve the accuracy and reliability of drone detection.

CHAPTER 3

SIMULATION SETUP AND METHODOLOGY

3.1 Introduction

This chapter provides a detailed exposition of the experimental framework designed to implement and evaluate the drone detection system. The primary challenge addressed in this simulation is the "small object detection" problem, where drones must be distinguished from birds and aircraft under varying environmental conditions[33]. The setup covers the entire pipeline, from hardware selection and software dependencies to advanced dataset preprocessing and the multi-model comparative training strategy.

3.2 Hardware Configuration

For deep learning tasks, the hardware must balance high-speed data throughput with parallel processing capabilities. The following specifications were utilized for the development and testing phases:

- **Processor:** Intel Core i5 / i7 or equivalent. High clock speeds are necessary for the initial data loading and the non-maximum suppression (NMS) calculations performed during the inference stage.
- **System Memory (RAM):** A minimum of 8 GB was utilized. This capacity ensures that high-resolution image batches (640x640) can be loaded into memory without causing bottlenecks during the training of deep neural networks.
- **Storage:** 256 GB or above SSD. Fast read/write speeds are essential for managing the dataset of 4,939 images and the various saved model weights (.pt files) generated during training.
- **Graphics Processing Unit (GPU):** NVIDIA GPU with CUDA support. The simulation relied on cloud-based Tesla T4/P100 GPUs via Google Colab to accelerate the tensor operations and backpropagation required by the YOLO architectures.
- **Camera Interface:** For real-time validation, a standard high-definition webcam was integrated to test the model's performance on live video streams.

3.3 Software Requirements and Development Environment

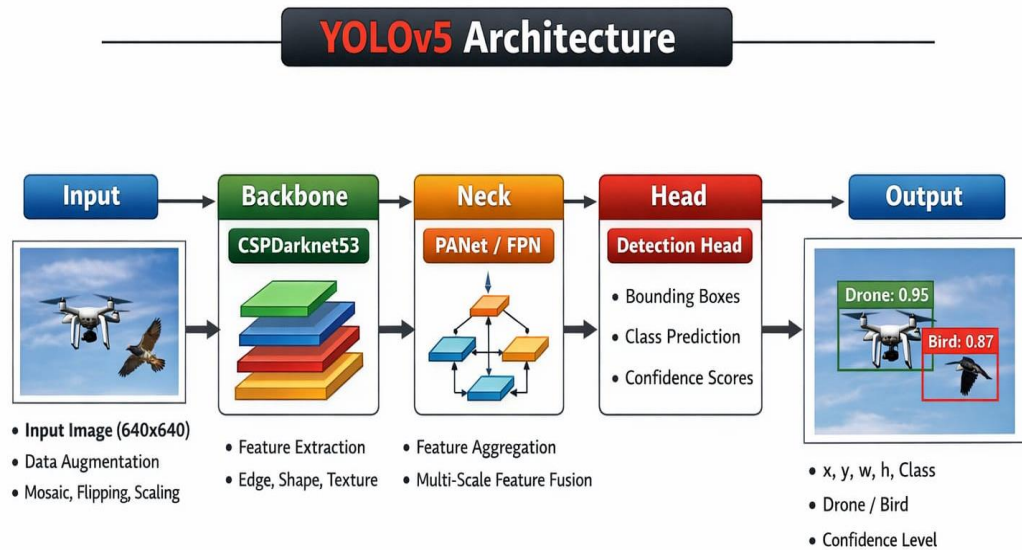


Fig 3.1 YoloV5 Architecture

- **Operating System:** A mix of Windows 11 for local development and Linux (Ubuntu) within the Google Colab environment.
- **Google colab Environment:** Google Colab is a cloud-based Python development platform that provides free GPU support for deep learning applications. In this project, it was used to train and evaluate the YOLOv5 model, perform dataset processing, and generate detection results efficiently.
- **Programming Language:** Python 3.x. The language choice was dictated by the availability of specialized libraries like PyTorch, NumPy, and OpenCV.
- **Deep Learning Framework:** PyTorch served as the primary framework, providing the computational graphs necessary for the YOLO (You Only Look Once) models.
- **Object Detection Architectures:** The simulation tested a range of architectures including YOLOv5, YOLOv8, YOLOv10, and YOLOv11, allowing for a study of the evolution of detection accuracy versus speed.
- **Annotation and Management:** Roboflow was the central platform for dataset versioning and labeling.

3.4 Dataset Preparation and Preprocessing

The robustness of the detection system is directly linked to the quality of the dataset.

3.4.1 Data Collection

A "Mixed Dataset" strategy was employed. Instead of relying on a single source, images were aggregated to include various drone types (quadcopters, fixed-wing), birds in flight, and aircraft. This prevents the model from developing a bias toward a specific aerial shape.

3.4.2 Annotation Process

Using Roboflow, each image was meticulously labeled with bounding boxes. Two primary classes were established:

- **Drone**
- **Bird**

This binary classification is critical for reducing false alarms in security contexts where birds are often misidentified as unauthorized drones.

3.4.3 Data Augmentation Techniques

To ensure the model can detect objects in low light, rain, or different angles, the following augmentations were applied to the training set:

- **Resizing:** All images were standardized to **640x640 pixels**.
- **Orientation:** Auto-orientation and horizontal flips.
- **Geometric Transformations:** Rotations between -15° and $+15^\circ$ and shear of $\pm 15^\circ$.
- **Color Space Adjustments:** Brightness variations (-25% to $+25\%$) and saturation adjustments to simulate different times of day.

3.4.4 Dataset Distribution

The fixed dataset of **4,939 images** was partitioned to ensure effective learning and unbiased evaluation:

- **Training Set (70%):** Used for the model to learn features and adjust weights.
- **Validation Set (20%):** Used during training to monitor for overfitting.
- **Testing Set (10%):** A "blind" set used only for the final performance report.

3.5 Model Configuration and Parameters

Each YOLO version was configured with specific hyperparameters to maintain a

consistent testing environment:

- **Batch Size:** 16/32 (depending on GPU memory).
- **Epochs:** 100 iterations to allow for the convergence of the loss function.
- **Learning Rate:** Optimized using the SGD (Stochastic Gradient Descent) or Adam optimizer.
- **Confidence Threshold:** Set at 0.25 for inference to ensure that low-probability detections are filtered out.
- **IoU Threshold:** Set at 0.45 for Non-Maximum Suppression (NMS) to handle overlapping objects.

3.6 Training Workflow

The training process followed a structured sequence:

- **Environment Setup:** Cloning the YOLO repository and installing dependencies (requirements.txt).
- **Data Integration:** Linking the Roboflow workspace via an API key to download the preprocessed dataset directly into the Colab environment.
- **Model Initialization:** Loading pre-trained weights (Transfer Learning) to speed up convergence.
- **Execution:** Running the training script while logging metrics like Box Loss, Class Loss, and Objectness Loss.

3.7 Datasets used

- In this project, multiple datasets were used to train and evaluate the drone detection model. Using different datasets helps improve the diversity of training samples and allows the model to learn various drone appearances under different environmental conditions. The datasets contain images of drones captured from different backgrounds, lighting conditions, and distances.
- Initially, a large dataset called Drone – V1 Fixed Dataset was used, which contains around 4939 images of drones. These images include drones in different orientations, scales, and backgrounds such as sky, buildings, and natural environments. This dataset helped the model learn general drone features and improve detection capability.
- To further analyze the performance of different YOLO models, smaller datasets

were also used. One of them is Drone Detection – V1 DRONE_ONEK, which contains 1000 drone images. This dataset was mainly used for comparing different YOLO models such as YOLOv5, YOLOv8, YOLOv10, and YOLOv11. By training each model on the same dataset, their detection performance could be evaluated under similar conditions.

- Another dataset used in this project is Drone – V1 DRONE_ONEKK, which contains 1085 images. This dataset includes drone images with varied angles, distances, and lighting conditions. It helped improve the model's ability to recognize drones in slightly complex scenarios.

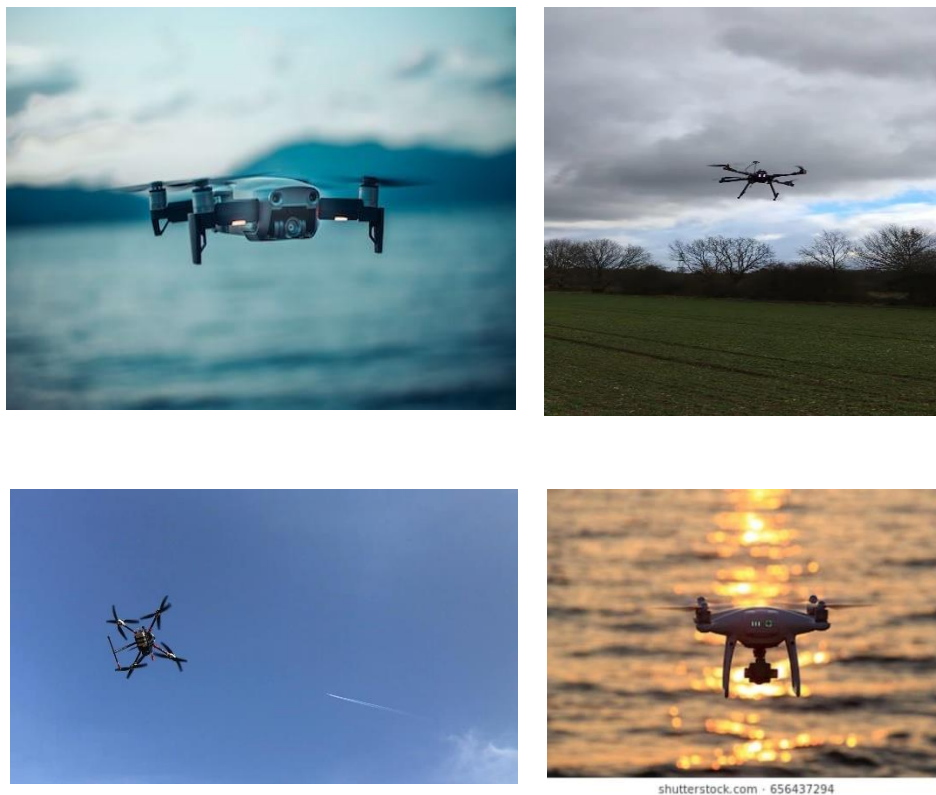


Fig 3.7.1 Sample Drone Datasets images

- In addition, the Drone detection 1, Drone detection 2 dataset was also used, which contains around 2044 images. This dataset includes a larger collection of drone images from multiple sources[34]. Combining this dataset with others increased the total number of training samples and improved the model's robustness.
- All datasets were carefully verified before training to ensure that the images and

labels were correct. Proper labelling is very important for object detection models because incorrect labels can reduce training performance. The datasets were also preprocessed and resized to the required input size for the YOLO model.

- By combining multiple datasets, the model was able to learn more diverse drone features and improve detection accuracy. This approach helped reduce overfitting and made the system more reliable for real-world drone detection applications..



Fig 3.7.1 Sample Drone images

3.8 Discussion of Simulation Observations

During the simulation, several key observations were made regarding the environment's impact on detection:

1. **Object Size:** Drones appearing smaller than 10x10 pixels in the 640x640 frame showed lower recall rates.
2. **Background Complexity:** Detection accuracy was significantly higher in clear sky backgrounds compared to complex urban or forested backgrounds.
3. **Model Evolution:** While newer models (v10/v11) offer architectural improvements, YOLOv5 demonstrated the highest precision and mAP_0.5 for this specific drone dataset, indicating its continued relevance for balanced datasets.

3.9 Summary

This chapter provided a comprehensive look at the simulation setup, highlighting the importance of the integrated hardware-software approach. By utilizing a diverse dataset of nearly 5,000 images and testing across multiple YOLO generations, the simulation established a high-standard benchmark for real-time.

CHAPTER 4

PROPOSED SYSTEM DESIGN

4.1 Introduction

The proposed system aims to develop an enhanced real-time drone detection system using deep learning techniques. With the rapid growth of drone usage in civilian and military applications, monitoring and detecting unauthorized drones has become an important security requirement. Traditional surveillance systems often fail to accurately detect drones due to limitations in distinguishing them from birds or other flying objects.

To overcome these limitations, this project proposes a deep learning based detection framework using the YOLOv5 object detection algorithm. The proposed system is designed to process images and video frames in real time and classify flying objects as either drones or birds. The system improves detection accuracy by combining multiple datasets and applying data preprocessing and augmentation techniques.

The proposed model follows a structured pipeline that includes data collection, image preprocessing, model training, object detection, and performance evaluation. The architecture ensures faster detection speed, improved classification accuracy, and reliable real-time monitoring capabilities.

4.2 System Architecture

The system architecture of the proposed enhanced drone detection system is designed to provide accurate and efficient identification of drones in real-time surveillance environments. The architecture integrates multiple components such as data acquisition, preprocessing, deep learning based object detection, and output visualization[35]. Each component plays an important role in the overall functioning of the system..

System Architecture

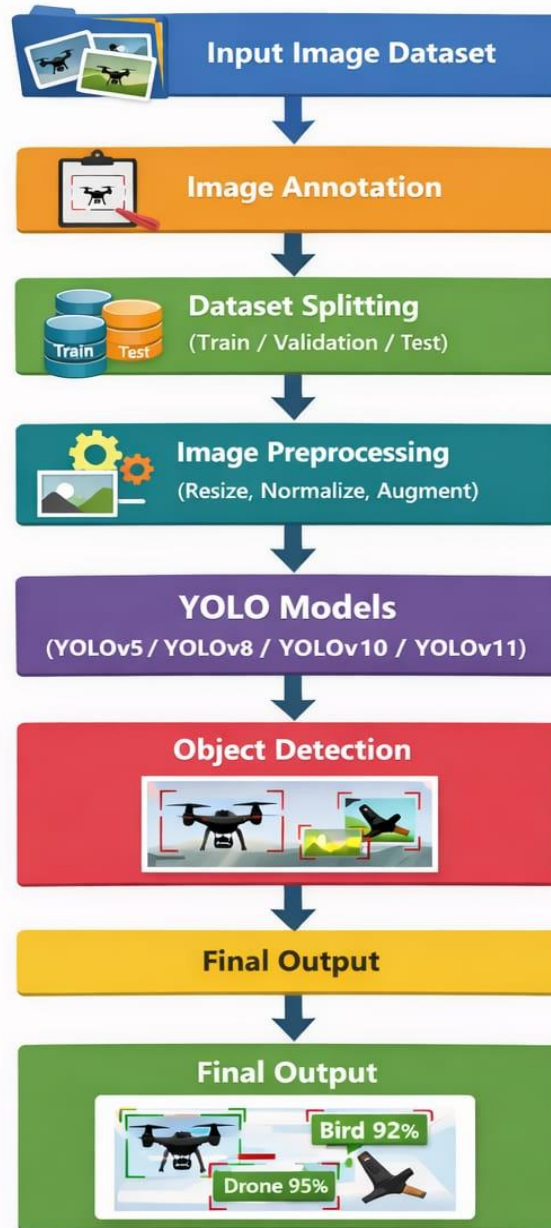


FIG 4.2 System Architecture

The architecture follows a sequential pipeline where input images are processed step-by-step until the final detection result is obtained. The major components of the system include image annotation, dataset splitting, image preprocessing, feature extraction, object detection using the YOLO model, classification, and output generation. This structured architecture ensures that the system can handle complex visual data and perform reliable

drone detection even under varying environmental conditions.

The first component of the system architecture is the data acquisition module. In this stage, the system collects images or video frames containing aerial objects such as drones and birds. The input data may come from publicly available datasets or from live camera feeds used for surveillance purposes. The collected dataset contains a variety of drone types, bird species, and background environments such as open sky, urban areas, and natural landscapes. Using a diverse dataset helps the model learn different object appearances and improves its ability to generalize to real-world scenarios.

After acquiring the dataset, the next stage of the architecture involves data annotation and dataset preparation. Each image in the dataset is carefully labeled using bounding boxes that identify the location of the object within the image[36]. The objects are classified into two categories: drone and bird. Annotation is an essential step because it provides the ground truth information required for training the object detection model. Proper labeling ensures that the model learns the correct visual patterns associated with each object class.

The annotated dataset is then divided into three subsets: training, validation, and testing datasets. The training dataset is used to train the deep learning model, allowing it to learn features from labeled images. The validation dataset helps monitor the model performance during training and prevents overfitting. The testing dataset is used after training to evaluate the final performance of the system.

The next stage of the system architecture is image preprocessing. Preprocessing is necessary to ensure that all input images are consistent and suitable for training the deep learning model. In this stage, images are resized to a standard resolution, typically 640×640 pixels, which matches the input size required by the YOLO model. Pixel values are normalized to improve training stability and faster convergence. In addition to these steps, data augmentation techniques such as rotation, flipping, scaling, and brightness adjustment are applied to increase dataset diversity. These techniques simulate different environmental conditions and improve the robustness of the model.

Once preprocessing is completed, the images are passed to the object detection model. The proposed system uses the YOLO (You Only Look Once) architecture, which is known for its high speed and accuracy in real-time object detection tasks. YOLO works by

processing the entire image in a single forward pass of the neural network, making it significantly faster than traditional object detection methods that rely on multiple processing stages.

The YOLO model consists of three main parts: backbone, neck, and detection head. The backbone network extracts important features from the input images using convolutional layers. These features represent patterns such as shapes, edges, and textures that help the model distinguish between drones and birds. The neck component combines features from different layers using feature pyramid networks to detect objects at multiple scales. This is particularly important for detecting small drones that appear far away in aerial images. Finally, the detection head predicts bounding boxes, confidence scores, and class probabilities for each detected object.

During the detection process, the model analyzes each image or video frame and identifies objects that match the learned patterns. If an object is detected, the system generates a bounding box around it and assigns a class label based on the predicted category. The system also calculates a confidence score indicating how certain the model is about the detection.

To remove duplicate detections, a technique called Non-Maximum Suppression (NMS) is applied. NMS eliminates overlapping bounding boxes and keeps only the most confident detection. This improves the clarity and accuracy of the detection results.

After object detection is completed, the system moves to the output visualization stage. In this stage, the detected objects are displayed on the screen with bounding boxes and class labels. For example, if a drone is detected in the image, a rectangular box is drawn around it and labeled as “Drone” along with the corresponding confidence score. This visual output allows users to easily identify drones present in the monitored area.

The final stage of the system architecture is performance evaluation. The effectiveness of the detection system is evaluated using standard performance metrics such as precision, recall, and mean Average Precision (mAP). Precision measures the accuracy of detected objects, while recall indicates the ability of the model to detect all relevant objects. The mAP metric provides an overall measure of detection accuracy across different confidence thresholds.

The proposed system architecture ensures efficient processing of visual data anaccurate

detection of drones in complex environments. By combining advanced deep learning techniques with proper dataset preparation and preprocessing, the system achieves improved detection accuracy and reduced false alarms. The architecture also supports real-time implementation, making it suitable for surveillance applications in airports, defense zones, and other sensitive areas.

The overall architecture of the proposed drone detection system consists of multiple stages that work together to detect and classify flying objects. The system architecture mainly includes the following components:

- Image Preprocessing
- Feature Extraction
- Object Detection using YOLOv5
- Classification of Drone and Bird
- Output Visualization
- Performance Evaluation

The system takes images or video frames as input, processes them using the trained deep learning model, and produces detection results showing the object class and its location within the frame.

4.3 Image annotation

Image annotation plays a crucial role in supervised learning, especially in object detection tasks like drone detection. It provides the ground truth information that the model uses to learn the relationship between input images and expected outputs. Without proper annotation, the model cannot accurately identify or localize objects in an image.

In this project, annotation was carefully performed to ensure high-quality labeling of drones and birds. Each object in the image was enclosed within a rectangular bounding box that tightly fits the object. This helps the model understand not only what the object is but also where it is located in the image. Special attention was given to avoid overlapping or incorrect bounding boxes, which can negatively impact training performance.

Additionally, consistency in labeling was maintained across the entire dataset. For example, drones were always labeled under a single class name (“Drone”) and birds under another (“Bird”). This avoids confusion during training and ensures that the model learns clear distinctions between different object categories.

Another important aspect of annotation is handling complex scenarios such as small objects, partially visible objects, and objects in motion. In aerial images, drones often appear very small or at different angles, so accurate annotation in such cases is essential[29]. Proper labeling in these challenging conditions improves the robustness of the model.

Furthermore, the annotated dataset was verified and corrected where necessary to eliminate errors. This quality checking process ensures that the dataset is reliable and suitable for training deep learning models like YOLO. High-quality annotation directly contributes to better accuracy, reduced false detections, and improved overall system performance.

4.4 Dataset Splitting:

Dataset splitting is an essential step in building a reliable and accurate deep learning model. It ensures that the model is trained, validated, and tested on different subsets of data, which helps in evaluating its real-world performance. Without proper dataset splitting, the model may simply memorize the training data instead of learning meaningful patterns.

In this project, the dataset was divided into training, validation, and testing sets using a standard ratio. The training set is the largest portion and is used to teach the model how to detect drones and birds by learning patterns, shapes, and features from annotated images. The validation set is used during the training process to monitor the model's performance and to fine-tune hyperparameters such as learning rate, batch size, and number of epochs.

The testing set plays a crucial role in evaluating the final performance of the model. It contains images that the model has never seen before. This helps in checking how well the model generalizes to new, unseen data. A good model should perform well not only on training data but also on validation and testing data.

Care was taken to ensure that the dataset was evenly distributed across all classes during splitting. This means that both drone and bird images were proportionally represented in training, validation, and testing sets. This balanced distribution prevents bias in the model and improves its ability to correctly classify different objects.

Another important aspect of dataset splitting is avoiding data leakage. Data leakage occurs when similar or duplicate images appear in both training and testing sets, which can

lead to overly optimistic performance results. In this project, proper measures were taken to ensure that each subset contains unique images.

Proper dataset splitting improves the robustness and reliability of the model. It helps in identifying overfitting and underfitting issues during training and ensures that the final system performs effectively in real-time detection scenarios.

4.5 Image Preprocessing:

Before feeding the images into the deep learning model, several preprocessing steps are applied to improve the quality and consistency of the dataset.

Image Resizing:

All images are resized to a fixed resolution of 640×640 pixels. This ensures compatibility with the input layer of the YOLOv5 model and maintains uniformity across the dataset.

Normalization

Pixel values are normalized to scale them within a specific range. This helps the neural network learn more efficiently and improves convergence during training.

Data Augmentation

To increase the diversity of the dataset and improve the model's generalization ability, various data augmentation techniques are applied. These include:

- Horizontal flipping
- Image rotation
- Scaling
- Brightness adjustment
- Contrast variation

These techniques help the model detect objects under different lighting conditions, orientations, and backgrounds.

4.6 YOLO models used

In this project, multiple YOLO (You Only Look Once) object detection models were explored and evaluated to identify the most suitable model for drone detection. The models considered include YOLOv5, YOLOv8, YOLOv10, and YOLOv11. Each of these models has its own architectural improvements and performance characteristics.

YOLO models are widely used in computer vision tasks due to their ability to perform real-time object detection with high accuracy. These models process the entire image in a single forward pass and predict bounding boxes along with class probabilities, making them efficient for surveillance applications.

During the simulation phase (as discussed in Chapter 3), all the selected YOLO models were trained using the same dataset consisting of drone and bird images. This ensured a fair comparison of their performance under identical conditions. The evaluation was carried out using standard metrics such as precision, recall, and mean Average Precision (mAP).

Each YOLO version has specific advantages:

YOLOv5 provides a good balance between speed and accuracy and is easy to implement using PyTorch.

YOLOv8 introduces improved architecture and better feature extraction capabilities.

YOLOv10 focuses on optimized performance and efficiency improvements.

YOLOv11 includes further enhancements in detection accuracy and model design.

After comparing all models, YOLOv5 was selected as the final model for the proposed system. This is because YOLOv5 demonstrated more consistent performance with higher precision and mAP values for the given drone detection dataset. It also provided stable results in distinguishing drones from birds and performed efficiently in real-time conditions.

Therefore, although multiple YOLO models were evaluated, the final implementation of the proposed system uses YOLOv5 for object detection due to its reliability, speed, and overall performance.

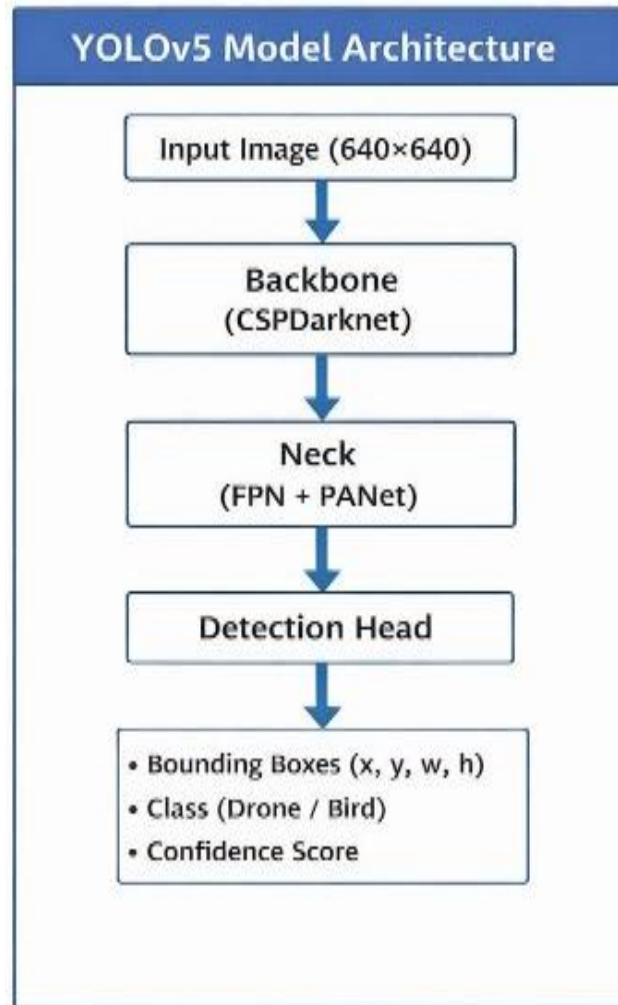


Fig 4.6 YoloV5 Model Architecture

4.7 Real-Time Object Detection

After training, the YOLOv5 model is used for real-time object detection using live camera input. The captured frames are preprocessed and passed to the model for detection. The system identifies objects and classifies them as drones or birds. Bounding boxes along with confidence scores are generated for each detected object. The final output is displayed in real time for monitoring purposes.

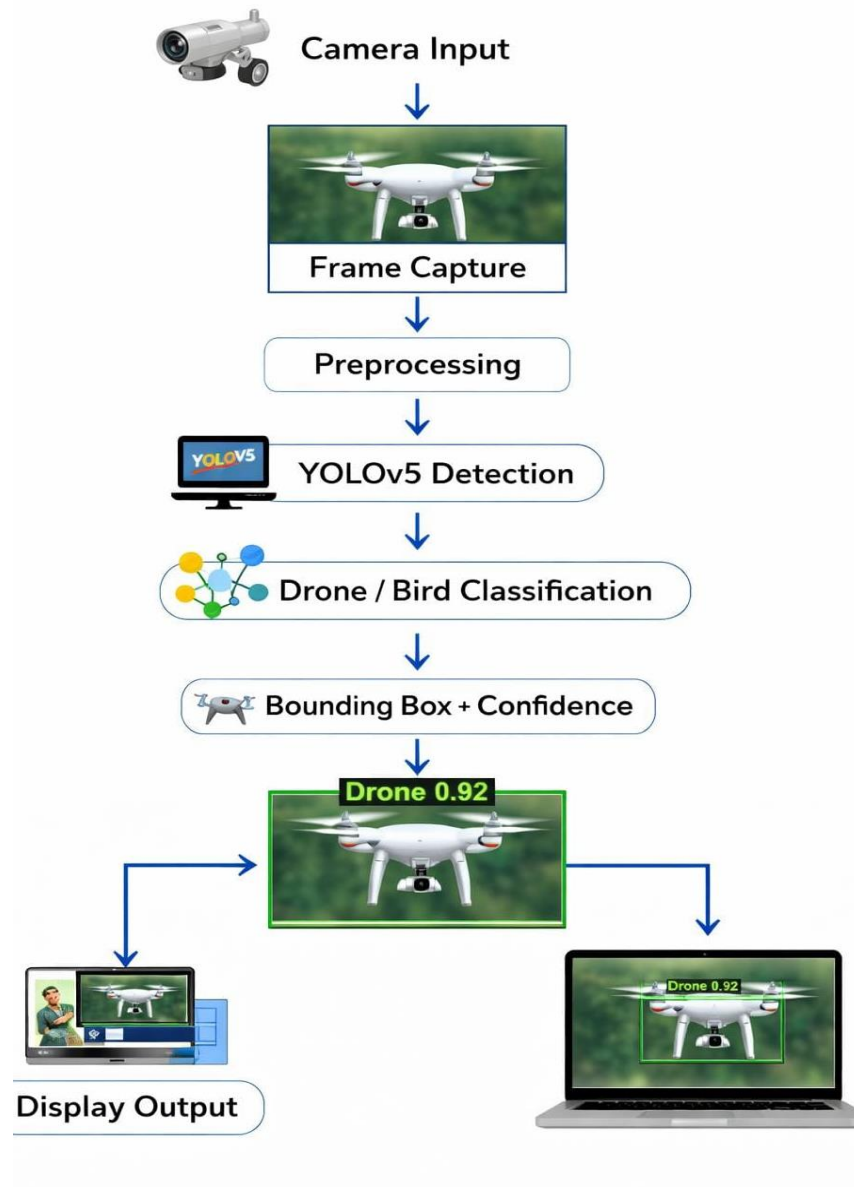


Figure 4.7 Real-Time Object Detection

The detection process includes the following steps:

- Capturing input frames from a camera
- Preprocessing the frames
- Passing the frames through the trained YOLOv5 model
- Detecting objects and predicting their class labels
- Drawing bounding boxes around detected objects
- Displaying the class label and confidence score

If the detected object is a drone, the system highlights it with a bounding box and displays the label “Drone” along with the detection confidence value.

4.8 Output Visualization

Output visualization is the final stage of the proposed drone detection system. In this stage, the detection results generated by the trained YOLOv5 model are displayed in a visual format so that users can easily understand and analyze the detected objects. Visualization plays an important role in surveillance systems because it provides clear information about the location and classification of objects detected in the input images or video frames.

After the real-time detection process is completed, the system produces the output by marking the detected objects with bounding boxes[37]. A bounding box is a rectangular box drawn around the object that indicates its exact position within the image. These boxes help users quickly identify the location of drones or birds in the frame.

Along with the bounding box, the system also displays a class label for each detected object. In this project, the model classifies objects into two categories: Drone and Bird. The predicted class label is shown near the bounding box so that users can easily understand what type of object has been detected.

Another important element shown in the output visualization is the confidence score. The confidence score represents the probability that the detected object actually belongs to the predicted class. It is usually displayed as a numerical value between 0 and 1 or as a percentage. A higher confidence score indicates that the model is more certain about its prediction.

For example, if the system detects a drone in an image, the output may display a bounding box around the object along with the label “Drone – 0.92”, which means the model is 92% confident that the object is a drone. This information helps users evaluate the reliability of the detection result.

The output visualization is generated using computer vision libraries such as OpenCV, which allows the system to draw bounding boxes, labels, and text directly on the image frames. The processed frames are then displayed on the screen or saved as image or video files for further analysis.

In the case of real-time video detection, the output frames are continuously updated as new frames are processed by the model. This creates a live detection interface where users can observe drones being detected instantly as they appear in the camera view. The system processes multiple frames per second, ensuring smooth and real-time visualization of detection results.

The visualization stage also makes it easier to evaluate the performance of the detection system. By observing the output frames, researchers can identify situations where the model performs well and cases where detection accuracy may decrease. This information can be used to further improve the dataset or adjust the training parameters.

Overall, output visualization provides a clear and user-friendly representation of the detection results produced by the deep learning model. It helps users quickly identify drones in surveillance environments and supports effective monitoring of restricted airspace. The visualization component therefore plays a critical role in transforming the detection results into meaningful information for practical security applications.

4.9 Performance Evaluation

Performance evaluation is an important stage in the proposed drone detection system. It helps to measure how effectively the trained YOLOv5 model detects and classifies objects such as drones and birds. Evaluating the performance of the system ensures that the detection model produces reliable and accurate results when applied to real-world surveillance scenarios.

In this project, the performance of the YOLOv5 object detection model is evaluated using several standard evaluation metrics commonly used in deep learning and computer vision tasks. These metrics help analyze different aspects of the model's performance, such as detection accuracy, classification reliability, and the ability to correctly identify objects in different conditions.

One of the primary metrics used in performance evaluation is Precision. Precision measures the proportion of correctly detected objects out of all objects predicted by the model. A high precision value indicates that the system produces fewer false detections, meaning that when the model predicts an object as a drone or bird, it is likely to be correct. Another important metric is Recall. Recall measures the ability of the model to detect all

relevant objects present in the dataset. It represents the ratio of correctly detected objects to the total number of actual objects present in the images. A high recall value indicates that the model is able to detect most of the drones and birds in the input images without missing them.

To balance both precision and recall, the F1-score is used as a combined performance metric. The F1-score is the harmonic mean of precision and recall and provides an overall measure of the model's detection performance. A higher F1-score indicates that the model performs well in both identifying objects and avoiding incorrect predictions.

Another key evaluation metric used in object detection is Mean Average Precision (mAP). This metric measures the overall accuracy of the object detection model by considering both the localization accuracy and classification performance. The mAP value is calculated by evaluating the precision and recall at different confidence thresholds. In object detection tasks, mAP provides a comprehensive understanding of how well the model detects objects across the entire dataset.

In addition to accuracy-related metrics, the system performance is also evaluated based on detection speed. Since the proposed system is designed for real-time drone detection, it is important to ensure that the model can process images quickly. The YOLOv5 model is capable of processing multiple frames per second (FPS), allowing the system to detect drones in real-time video streams without significant delay.

The evaluation process involves testing the trained model on a separate validation dataset that was not used during training. This helps measure how well the model generalizes to new and unseen data. The validation dataset contains images of drones and birds under different environmental conditions, such as varying lighting, backgrounds, and object sizes.

During the evaluation phase, the detection results are compared with the ground truth annotations present in the dataset. Ground truth annotations represent the actual object locations and class labels defined during the dataset labeling process. By comparing predicted results with ground truth data, the system calculates the performance metrics accurately.

The performance evaluation results help identify the strengths and limitations of the proposed detection system. For example, the system may perform very well in detecting

drones in clear sky backgrounds but may face challenges when objects appear extremely small or when the background is highly complex. Such observations can guide further improvements in dataset quality, training parameters, or model configuration.

Overall, the performance evaluation process confirms the effectiveness of the YOLOv5-based drone detection model. The evaluation metrics demonstrate that the system can accurately detect drones while distinguishing them from birds, thereby reducing false alarms in surveillance applications. This makes the proposed system reliable for real-time monitoring and security purposes.

4.10 Advantages of Proposed System

The proposed drone detection system offers several advantages compared to traditional detection methods.

- High detection accuracy using deep learning
- Ability to distinguish drones from birds
- Faster real-time object detection
- Improved performance in complex environments
- Scalable system that can be extended for real-time surveillance
- Efficient detection of small objects: The system can detect small drones that appear far away in aerial images due to the multi-scale detection capability of the YOLOv5 model.
- Reduced false alarms: By training the model with both drone and bird datasets, the system minimizes incorrect detections and improves reliability.
- Real-time monitoring capability: The system can process video frames continuously, allowing security personnel to monitor restricted areas without delay.
- User-friendly visualization: The detected objects are displayed with bounding boxes and class labels, making it easier for users to understand the detection results.
- Flexibility and adaptability: The system can be trained with new datasets to detect additional aerial objects such as helicopters, airplanes, or other flying vehicles.
- Cost-effective implementation: The proposed system can run on standard computers with GPU support, reducing the need for expensive surveillance equipment.
- Easy integration with security systems: The detection model can be integrated with

existing surveillance cameras and monitoring systems for enhanced security applications.

- Support for automated surveillance: The system reduces the need for continuous human monitoring by automatically detecting drones in the surveillance area.

4.11 Summary

This chapter presented the detailed design and implementation of the proposed drone detection system. The proposed system is developed using deep learning techniques with the YOLOv5 object detection model to accurately detect drones and distinguish them from birds in aerial images and video streams. The chapter explained the overall workflow and different components involved in building the detection system.

Initially, the system architecture was described to illustrate how the different modules of the proposed system interact with each other. The architecture includes various stages such as input data acquisition, preprocessing of images, model training, object detection, and output visualization. Each stage plays a significant role in ensuring the proper functioning of the system.

The chapter also discussed the process of input data acquisition, where a dataset consisting of drone and bird images was collected from multiple sources. The dataset contains images captured under different environmental conditions and backgrounds to ensure that the model can perform effectively in real-world situations. Proper dataset organization and annotation were also explained as essential steps for training the object detection model.

Next, the image preprocessing stage was explained, where input images are prepared before being fed into the deep learning model. Preprocessing includes resizing images, normalizing pixel values, and organizing the dataset in a suitable format for training. These steps help improve the efficiency and accuracy of the model during the learning process.

The chapter further described the YOLOv5 object detection model, which forms the core of the proposed system. The architecture of YOLOv5, including the backbone, neck, and detection head components, was discussed in detail. The ability of YOLOv5 to detect objects quickly and accurately makes it suitable for real-time drone detection applications.

In addition, the real-time object detection process was explained, where the trained

model analyzes input images or video frames to identify and classify flying objects. The system processes each frame and predicts bounding boxes along with class labels for detected objects. Techniques such as Non-Maximum Suppression are used to remove duplicate detections and improve result accuracy.

The output visualization stage was also described, where the detection results are displayed by drawing bounding boxes and class labels on the detected objects. This visual representation allows users to easily identify drones and birds in the surveillance environment.

Furthermore, the performance evaluation of the proposed system was discussed using standard evaluation metrics such as precision, recall, F1-score, and mean average precision. These metrics help measure the accuracy and effectiveness of the detection model. The system was also evaluated based on detection speed to ensure that it meets real-time monitoring requirements.

The chapter also highlighted the advantages of the proposed system, including high detection accuracy, real-time processing capability, reduced false alarms, and the ability to detect small aerial objects. These advantages make the system suitable for applications such as airport monitoring, border security, and surveillance systems.

Overall, this chapter provided a comprehensive explanation of the proposed system design and implementation. The combination of deep learning techniques and the YOLOv5 object detection model enables the system to detect drones efficiently and accurately. The detailed discussion of system processes establishes the foundation for analyzing the experimental results and performance evaluation presented in the next chapter.

CHAPTER 5

RESULTS AND DISCUSSIONS

5.1 Introduction to the Evaluation

This Chapter presents the experimental results and analytical discussion of the drone detection system developed using YOLO-based object detection models. The objective of the experimentation phase was to evaluate different versions of the YOLO architecture and determine the most suitable model for accurate drone detection across multiple datasets. The evaluation also included a cross-dataset generalization study and the application of an image enhancement technique, namely the Rolling Guidance Filter (RGF), to investigate its effect on model performance.

Drone detection is an important task in many applications such as surveillance, airspace monitoring, defense systems, and security-critical infrastructure protection. However, detecting drones in real-world environments presents several challenges. Drones are typically small objects in images, often appear at varying distances, and may be affected by lighting conditions, motion blur, or background clutter. These factors make reliable detection difficult for computer vision systems.

To address these challenges, this project evaluates multiple YOLO models trained on the same dataset. The goal of this initial experiment was to identify which YOLO architecture provides the best detection accuracy and stability for the given drone dataset. After determining the best performing model, further experiments were conducted using cross-dataset evaluation in order to test the robustness and generalization ability of the selected model.

In addition, image enhancement was applied to the cross datasets using the Rolling Guidance Filter in MATLAB. This enhancement technique was used to improve image clarity and edge preservation while reducing noise. The performance of the YOLOv5 model was then evaluated on both the original cross datasets and the enhanced datasets. The results were compared in order to determine whether image enhancement improves detection accuracy.

The following subsections present the results obtained during each stage of experimentation and provide a detailed discussion of the observed performance trends.

5.2 Comparison of Different YOLO Models

In the first stage of the project, multiple YOLO models were trained and evaluated using the same drone dataset. The models considered in this experiment included YOLOv5, YOLOv8, and YOLOv11. The purpose of this comparison was to determine which architecture performs best for the specific task of drone detection.

Each model was trained using identical training conditions, including the same training dataset, number of epochs, batch size, and evaluation metrics. This ensured that the comparison remained fair and that any differences in performance could be attributed to the architectural differences between the models rather than variations in training conditions.

The performance of the models was evaluated using standard object detection metrics such as Precision, Recall, F1-Score, and Mean Average Precision (mAP). These metrics provide a comprehensive understanding of the detection capability of each model. Precision measures the correctness of predicted detections, while Recall measures the model's ability to detect all drones present in the image. The mAP score provides an overall indicator of detection performance across different confidence thresholds.

TABLE 5.2.1 : Comparison of different models by using Drone dataset1

MODELS	PRECISION	RECALL	MAP_0.5	MAP_0.5:0.9
YOLOV5	0.89182	0.85933	0.9012	0.59847
YOLOV8	0.86999	0.82505	0.86547	0.60929
YOLOV10	0.83316	0.76458	0.84214	0.59598
YOLOV11	0.87653	0.80216	0.85843	0.59837

The results presented in Table 5.2 demonstrate a comparative analysis of different YOLO architectures for drone detection. Among the evaluated models, YOLOv5 achieved the highest Precision (0.89182) and mAP@0.5 (0.9012), indicating its superior ability to accurately detect drones with minimal false positives.

5.2.1 GRAPHS:

YOLOV5:

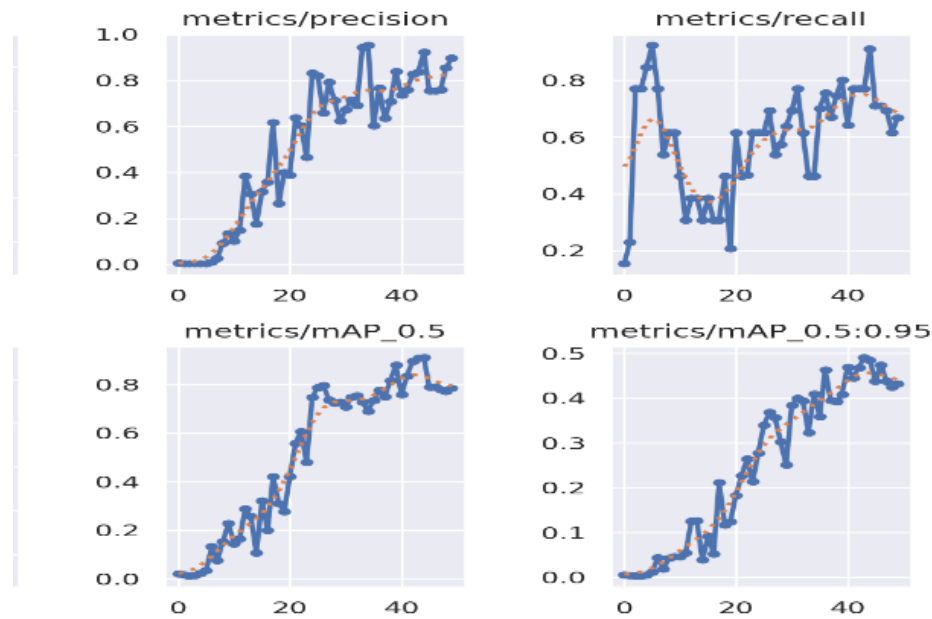


Figure 5.2.1 Performance metrics(precision, recall, mAP@0.5 and mAP@0.5:0.9) for YoloV5

The graphs in Figure 5.2.1 represent the training performance of the YOLOv5 model in terms of precision, recall, mAP@0.5, and mAP@0.5:0.95. A steady increase in these metrics indicates effective learning and good convergence behavior of the model. Precision shows the accuracy of positive predictions, while recall measures the model's ability to detect all actual objects. The mean Average Precision (mAP) metrics evaluate overall detection performance at different IoU thresholds, with mAP@0.5 being a standard measure and mAP@0.5:0.95 providing a stricter, averaged assessment across IoU values. The upward trends in all four plots confirm that the YOLOv5 model is training successfully and improving its object detection capability.

In addition to the observed trends, minor fluctuations in the curves can be attributed to variations in training data and batch updates during optimization. Despite these variations, the overall performance remains stable and consistent across epochs. The improvement in mAP@0.5:0.95 highlights the model's robustness under stricter evaluation conditions. Furthermore, the balance between precision and recall indicates that the model effectively minimizes both false positives and false negatives.

YOLOV8:

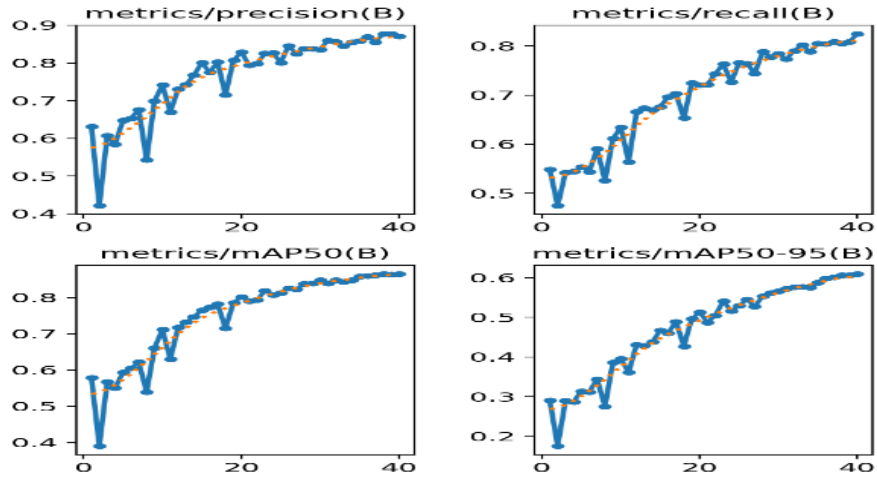


Figure 5.2.2 Performance metrics(precision, recall, mAp@0.5 and mAP@0.5:0.9) for YoloV8

The graph illustrates the performance metrics of the YOLOv8 model during training. The gradual improvement in precision, recall, and mAP values indicates stable learning, though slightly lower compared to YOLOv5.

YOLOV10:

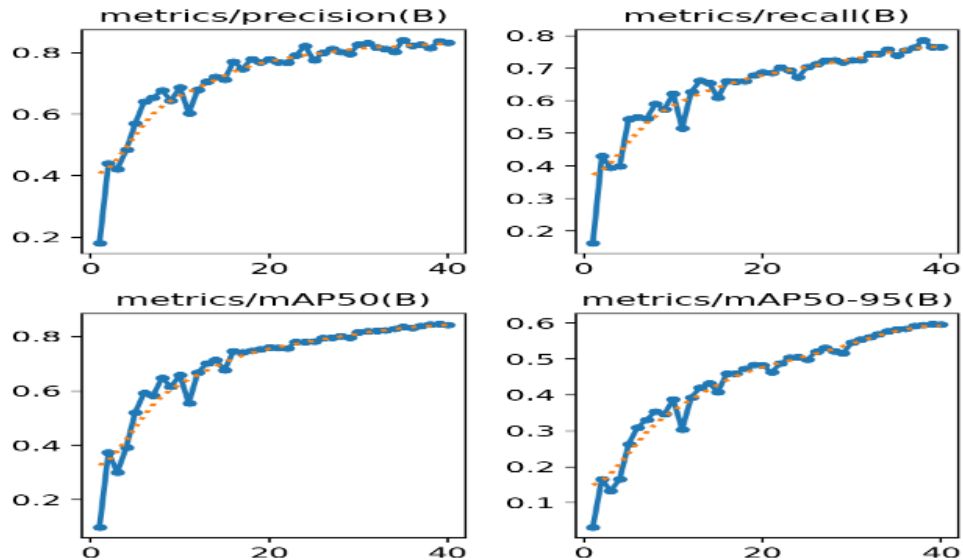


Figure 5.2.3 Performance metrics(precision, recall, mAp@0.5 and mAP@0.5:0.9) for YoloV10

The graph shows the training behavior of the YOLOv10 model. Although the metrics improve over epochs, the performance is comparatively lower, indicating reduced detection efficiency for this dataset.

YOLOV11:

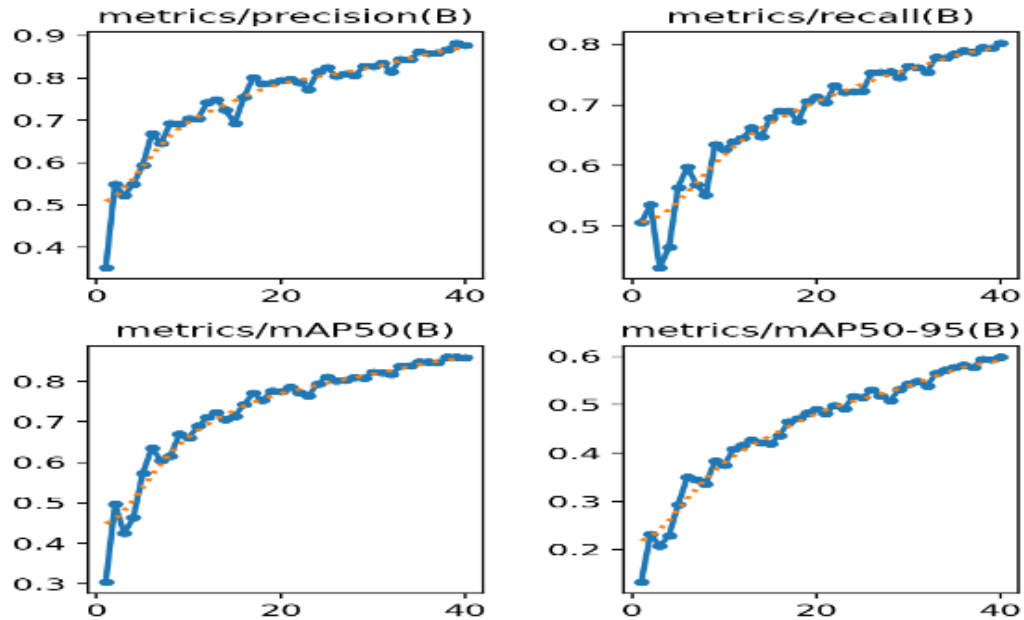


Figure 5.2.4 Performance metrics(precision, recall, mAp@0.5 and mAp@0.5:0.9) for YoloV11

The graph represents the training performance of the YOLOv11 model. The model shows consistent improvement in metrics, demonstrating stable learning with moderate detection accuracy.

Additionally, YOLOv5 also recorded the highest Recall (0.85933), showing its effectiveness in identifying a larger number of true drone instances in the dataset.

YOLOv8, while slightly lower in Precision and Recall compared to YOLOv5, achieved the highest mAP@0.5:0.9 score (0.60929), suggesting better performance across stricter IoU thresholds and improved localization capability. YOLOv11 demonstrated competitive Precision (0.87653) but relatively lower Recall (0.80216), indicating a trade-off between detection accuracy and completeness. On the other hand, YOLOv10 showed comparatively lower performance across all evaluation metrics, with reduced Precision (0.83316) and Recall (0.76458), making it less effective for this specific task.

Overall, the comparison indicates that YOLOv5 is the most suitable model for drone detection in this study due to its balanced and superior performance across key metrics.

5.3 TRAINING YOLOV5 ON MULTIPLE DRONE DATASETS

After selecting YOLOv5 as the best performing model, the next stage of the project involved training YOLOv5 on multiple drone datasets. The purpose of this experiment was to evaluate how well the model performs when exposed to different drone image distributions.

Drone datasets often vary in terms of image resolution, environmental conditions, drone types, camera angles, and background complexity. These variations can significantly influence the performance of object detection models. Therefore, evaluating the model on multiple datasets helps determine whether the detection system is robust and adaptable to different scenarios.

In this experiment, several drone datasets were collected and prepared for training. The datasets contained labeled images with bounding box annotations indicating the location of drones in each image. Before training, the datasets were converted into a format compatible with YOLOv5 training. The YOLOv5 model was trained separately on each dataset using the same training configuration. The resulting performance metrics were then recorded and analyzed. The experiments showed that YOLOv5 maintained strong detection performance across multiple datasets. However, minor variations in accuracy were observed depending on the characteristics of each dataset.

TABLE 5.3.1 : TRAINED VALUES OF MULTIPLE DRONE DATASETS USING YOLOV5 MODEL

DATASETS	PRECISION	RECALL	mAP_0.5	mAP_0.5:0.9
Drone dataset 1	0.97952	0.97112	0.99355	0.78129
Drone dataset 2	0.9952	0.9726	0.98653	0.80831
Drone dataset 3	0.93827	0.90588	0.9326	0.52134
Drone dataset 4	0.89182	0.85933	0.9012	0.59847

The results presented in Table 5.3.1Table highlight the performance of the YOLOv5 model when trained on multiple drone datasets of varying sizes and characteristics. It can be observed that the model achieves higher Precision, Recall, and mAP values for smaller and more focused datasets such as Drone-one-k and Drone-one-kk. This indicates that the model performs exceptionally well when the dataset contains more consistent and less complex image variations.



Fig 5.2 Detected images of Different Yolo Models

As the dataset size increases and becomes more diverse, such as in Drone Dataset 1 and the Fixed dataset, a slight decrease in performance metrics can be observed. This reduction can be attributed to increased variations in image conditions, including background complexity, lighting variations, and drone scale differences. Despite this, YOLOv5 maintains stable and reasonably high performance across all datasets, demonstrating its robustness and generalization capability.

Furthermore, the $mAP@0.5:0.9$ values show a noticeable variation across datasets, indicating that stricter IoU thresholds pose additional challenges for precise localization. However, the model continues to perform consistently, proving its effectiveness in real-world detection scenarios.

To better understand the training behavior and convergence of the model, graphical representations of key performance metrics such as Precision, Recall, and mAP across training epochs are presented in the following section.

5.3.1 GRAPHS

Drone dataset 1:

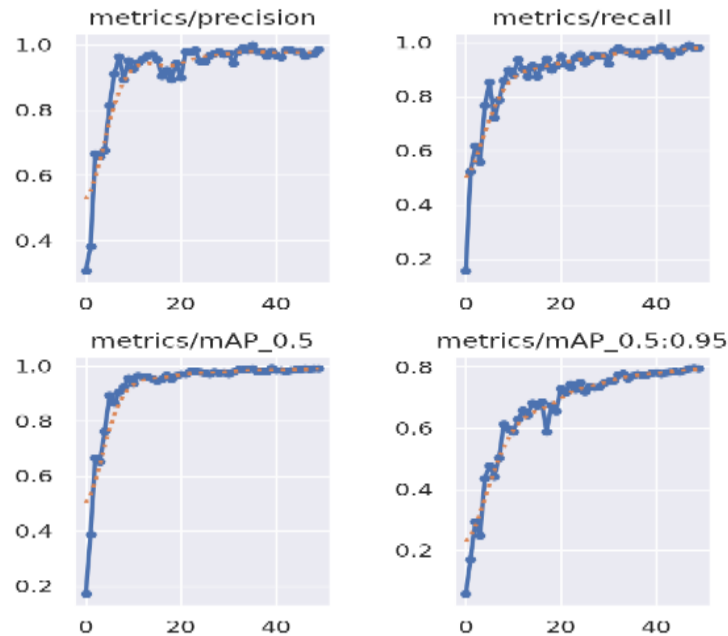


Figure 5.3.1 Performance metrics(precision, recall, mAp@0.5 and mAP@0.5:0.9) of Drone detection 1

The graph illustrates the training performance of the YOLOv5 model on the Drone-v1 dataset. It can be observed that precision, recall, and mAP values increase steadily during the initial epochs, indicating effective learning of features. The curves gradually stabilize as training progresses, showing proper convergence of the model. The high values of evaluation metrics reflect strong detection accuracy and reliable performance on this dataset.

The rapid rise in performance metrics during the early epochs indicates efficient feature extraction by the model. The consistency in precision and recall values suggests a balanced detection capability with minimal false predictions. The improvement in mAP@0.5:0.95 further demonstrates the model's ability to perform well under stricter evaluation criteria.

Drone dataset 2:

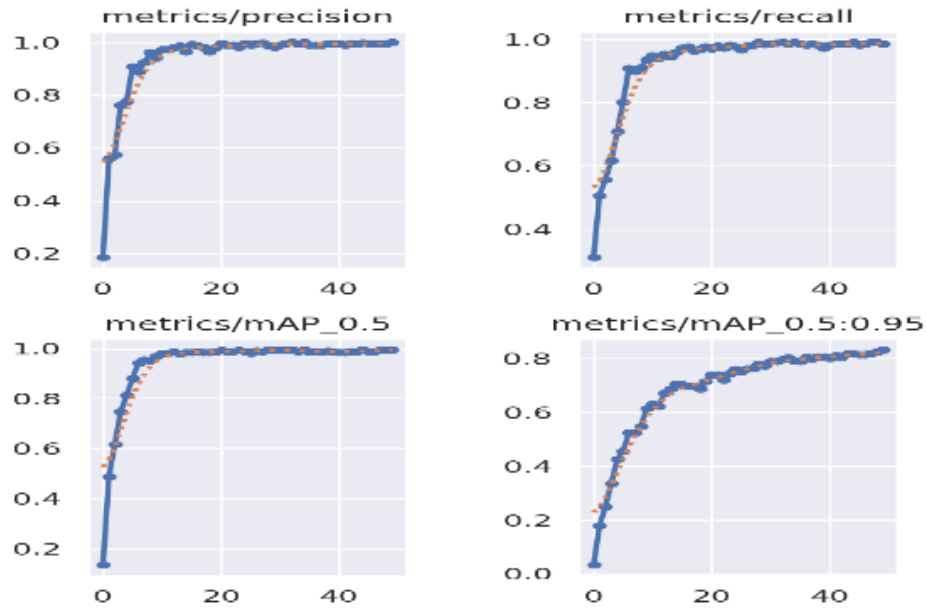


Figure 5.3.2 Performance metrics(precision, recall, mAp@0.5 and mAP@0.5:0.9) of Drone dataset 2

The graph represents the training behavior of the YOLOv5 model on the Drone-v1 dataset. The consistent improvement in precision, recall, and mAP values indicates stable learning and efficient feature extraction. The smooth convergence of the curves suggests that the model performs well without significant fluctuations. Overall, the model achieves good detection performance on this dataset.

The high and stable precision and recall values demonstrate the model's ability to accurately detect objects while minimizing errors. The steady improvement in mAP@0.5 and mAP@0.5:0.95 reflects strong generalization performance under varying IoU thresholds. Minor variations observed are negligible and do not impact overall model reliability.

Drone dataset 3:

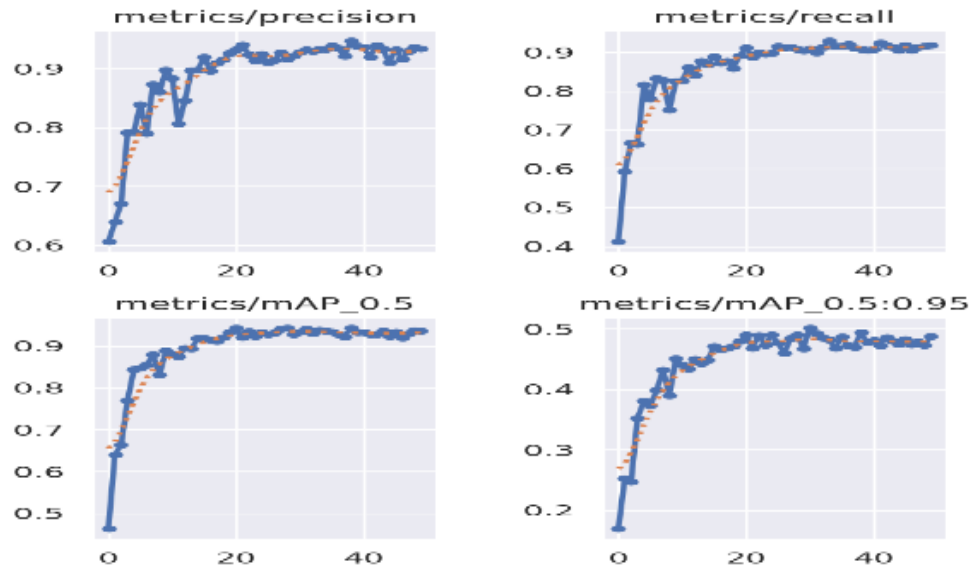


Figure5.3.3 Performance metrics(precision, recall, mAp@0.5 and mAP@0.5:0.9) of Drone dataset 3

The graph shows the training performance of the YOLOv5 model on the combined on the All Drone dataset 3 dataset. Although the metrics improve steadily over epochs, slight variations can be observed due to increased dataset diversity and complexity. The model demonstrates stable learning and reasonable convergence. Overall, the results indicate that YOLOv5 is capable of handling diverse drone data with consistent performance.

The observed trends indicate that the model effectively adapts to the increased diversity present in the combined dataset. The gradual improvement in precision and recall highlights the model's ability to accurately identify objects even in more complex scenarios. The mAP@0.5 values show strong detection performance, while the steady rise in mAP@0.5:0.95 reflects improved localization accuracy under stricter IoU thresholds. Slight fluctuations in the curves are expected due to variations in object scale, background, and dataset complexity. Despite these variations, the model maintains consistent performance across training epochs. The convergence of all metrics suggests that the model is well-trained and avoids overfitting. The balanced precision and recall values indicate

effective handling of both false positives and false negatives.

Drone dataset 4:

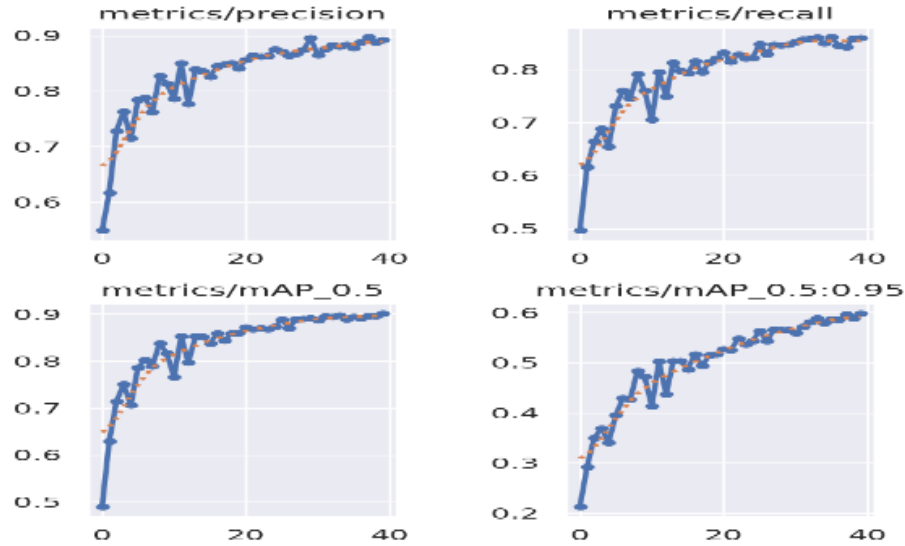


Figure 5.3.4 Performance metrics(precision, recall, mAp@0.5 and mAP@0.5:0.9) of Drone dataset 4

The graphical results presented above illustrate the training progression of the YOLOv5 model across different drone datasets. It can be observed that, for all datasets, the Precision, Recall, and mAP curves show a steady increase during the initial epochs, followed by gradual stabilization as training progresses. This indicates that the model effectively learns the underlying features and converges properly without significant fluctuations.

Datasets such as Drone dataset 1 and Drone dataset 2 demonstrate faster convergence and higher stability, which can be attributed to more uniform data distribution and lower complexity. In contrast, Drone dataset 3 and the Drone dataset 4 exhibit comparatively slower convergence and minor fluctuations in the curves, reflecting the increased diversity and complexity of these datasets.

Despite these variations, all graphs indicate consistent learning behavior and reliable performance of the YOLOv5 model. The smooth trends in the evaluation metrics confirm that the model is neither underfitting nor overfitting significantly, thereby ensuring good generalization capability. Datasets containing clearer drone images with less background clutter generally produced higher detection accuracy. In contrast, datasets containing small drone objects, complex backgrounds, or low-resolution images resulted in slightly lower

detection performance. Despite these variations, YOLOv5 demonstrated strong adaptability across different data sources.

5.4 ENHANCEMENT OF CROSS – DATASETS:

All drone datasets were processed using the Rolling Guidance Filter (RGF) in MATLAB before training the YOLOv5 model.

The Rolling Guidance Filter is an edge-preserving smoothing technique that effectively removes noise and small-scale textures while maintaining important structural details such as object boundaries.. The enhanced datasets were then used to retrain the YOLOv5 model using the same training configuration as in previous experiments. Performance metrics such as Precision, Recall, F1-Score, and MAP were recorded and compared with the results obtained from the original (non-enhanced) datasets.



Fig 5.4.1 Detected drone images of Enhanced Drone- Dataset1



Fig 5.4.2 Detected drone images of Enhanced Drone – Dataset2

The results indicated that image enhancement using the Rolling Guidance Filter led to improved detection performance in most cases. Specifically, there was a noticeable

improvement in Precision and MAP values, suggesting that the model was able to better distinguish drones from the background. Additionally, smoother training curves were observed in the graphical analysis, indicating more stable convergence during training.

However, in some cases, only marginal improvements were observed, which may be due to the already high quality of certain datasets or the loss of fine texture details during filtering. Despite this, the overall results demonstrate that preprocessing techniques like Rolling Guidance Filter can enhance model robustness and improve detection performance, especially in challenging visual conditions.

TABEL 5.4.1 : ENHANCED DATASET VALUES

Datasets-names	Precision	Recall	mAP@0.5	mAP@0.5:0.9
Drone-Dataset1	0.897	0.874	0.879	0.444
Drone-Dataset2	0.95	0.912	0.939	0.657

The results presented in Table 5.4.1 illustrate the performance of the YOLOv5 model on enhanced datasets using the Rolling Guidance Filter. When compared with the original cross-dataset results, it can be observed that image enhancement has led to noticeable improvements in several performance metrics.

For the Drone-Dataset1, there is a slight improvement in detection stability, although minor variations in Precision and Recall are observed. In the Drone- Dataset2, the model shows improved Precision and mAP values, indicating better discrimination between drone objects and background after enhancement. Similarly, Dataset1 demonstrates improved mAP@0.5:0.9, suggesting enhanced localization performance under stricter evaluation conditions.

Overall, the enhancement process helps in reducing background noise and improving feature clarity, allowing the model to learn more relevant features during training. This results in better generalization and improved detection performance across unseen data. To further validate these observations, graphical analysis using Precision-Recall curves, Confidence curves, and F1-score curves is presented in the following section.

5.5 GRAPHS:(ENHANCED)

Drone-Dataset1

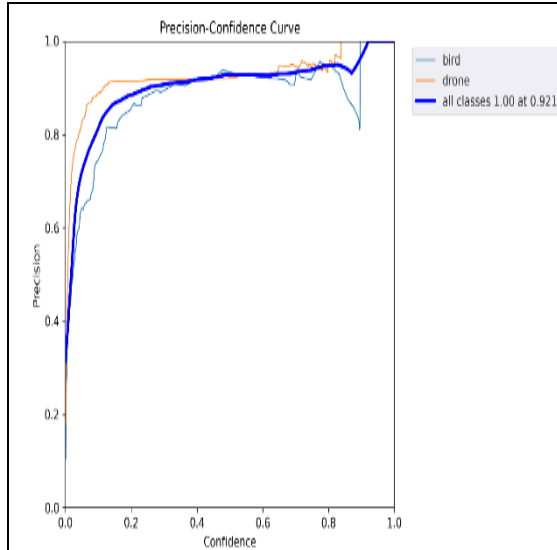


Fig 5.4.3 Precision-confidence curve showing variation of precision with confidence threshold.

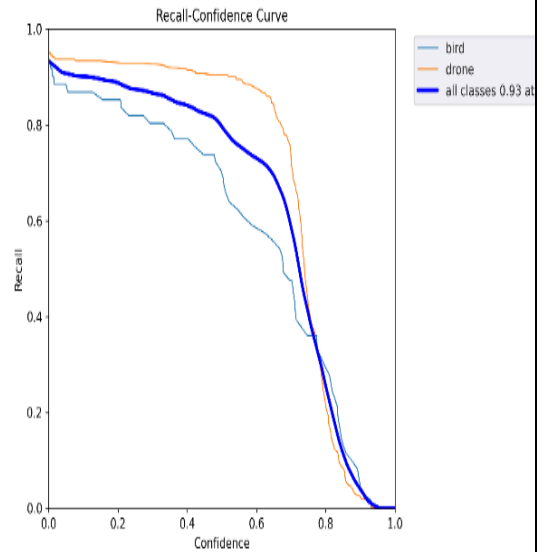


Fig 5.4.4 Recall-confidence curve showing variation of recall with confidence threshold

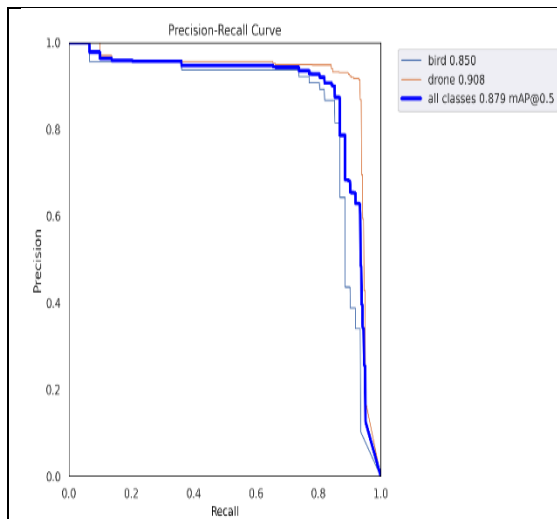


Fig 5.4.5 Precision-Recall confidence curve showing variation of precision with recall threshold

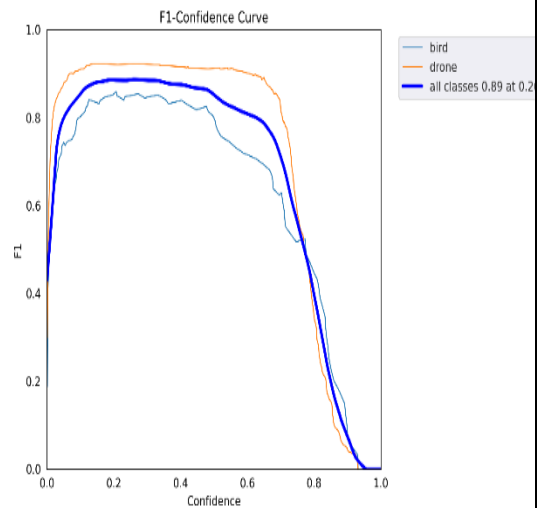


Fig 5.4.6 F1-score-confidence curve showing variation of F1-score with respect to confidence threshold.

Drone-Dataset2:

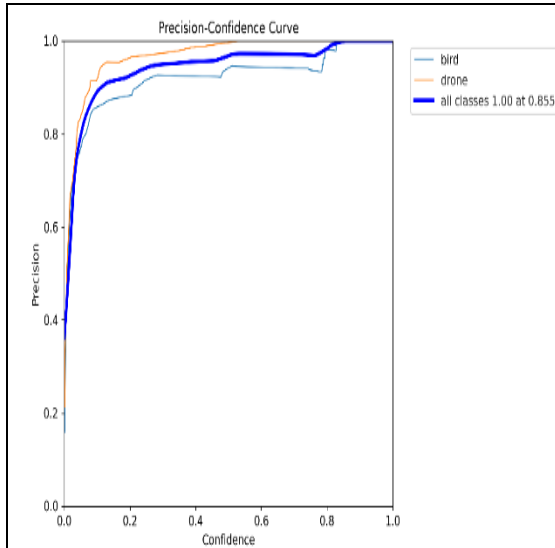


Fig 5.4.7 Precision–confidence curve showing variation of precision with confidence threshold.

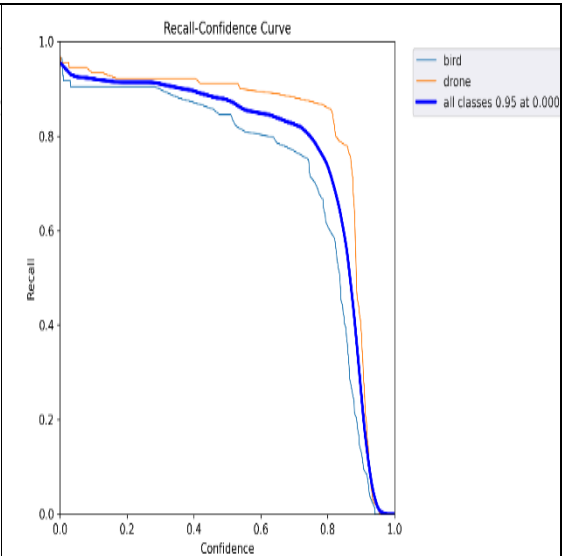


Fig 5.4.8 Recall–confidence curve showing variation of recall with confidence threshold

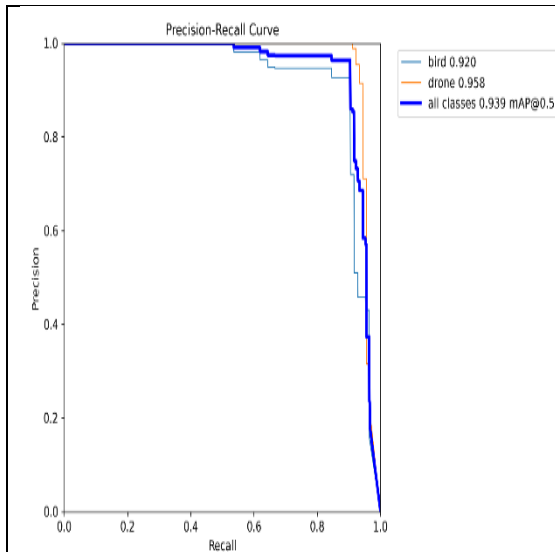


Fig 5.4.9 Precision–Recall confidence curve showing variation of precision with recall threshold

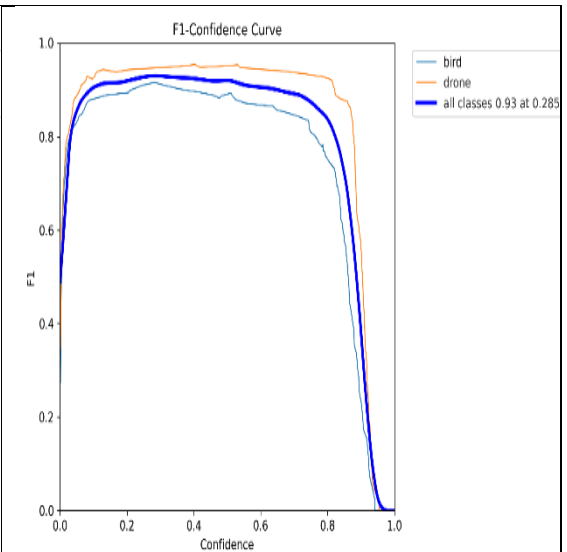


Fig 5.4.10 F1-score–confidence curve showing variation of F1-score with respect to confidence threshold.

The graphical results further support the improvements observed in the tabular data. The Precision-Recall curves show a better balance between precision and recall after

enhancement. The Confidence curves indicate more stable predictions across varying thresholds, while the F1-score curves demonstrate improved overall performance.

Additionally, smoother and more consistent curves suggest that the model achieves better convergence when trained on enhanced datasets. This indicates that preprocessing using the Rolling Guidance Filter contributes to more stable learning and improved robustness. Overall, the combination of tabular and graphical analysis confirms that image enhancement plays a significant role in improving the performance of YOLOv5 for drone detection tasks, especially in complex and noisy environments.

5.5 COMPARATIVE ANALYSIS: ORIGINAL vs ENHANCED DATASETS

To evaluate the impact of image enhancement on model performance, a comparative analysis was conducted between the original cross-dataset results and the enhanced datasets processed using the Rolling Guidance Filter (RGF). The comparison focuses on key performance metrics including Precision, Recall, mAP@0.5, and mAP@0.5:0.9.

The results clearly indicate that image enhancement contributes to improved detection performance across most datasets. The Rolling Guidance Filter helps in reducing noise and preserving important structural features, allowing the model to better distinguish drone objects from complex backgrounds.

TABLE 5.5.1 : Comparison of Original and Enhanced Datasets

Datasets-names	Type	Precision	Recall	Map_0.5	Map_0.5:0.9
Drone-Dataset1	Original	0.800	0.771	0.740	0.383
	Enhanced	0.897	0.874	0.939	0.444
Drone-Dataset2	Original	0.937	0.867	0.917	0.657
	Enhanced	0.950	0.912	0.939	0.658

From Table 5.5.1, it can be observed that the application of the Rolling Guidance Filter leads to noticeable improvements in detection performance for most datasets.

For the Drone-Dataset1, there is a significant improvement across all metrics. Precision increased from 0.800 to 0.897, and mAP@0.5 improved from 0.740 to 0.879. This indicates

that enhancement effectively reduces background noise and improves feature clarity, allowing the model to detect drones more accurately.

In the Drone-Dataset2, moderate improvements are observed. Precision and Recall show slight increases, while MAP values also improve marginally. This suggests that enhancement benefits datasets with moderate complexity, although improvements may be limited when the original dataset is already well-structured.

Overall, the comparison demonstrates that image enhancement using the Rolling Guidance Filter improves model performance in noisy and complex environments but may introduce minor trade-offs in already optimized datasets.

The graphical comparisons for each performance metric are shown below.

5.5.1 COMPARISON GRAPHS:(BEFORE VS AFTER ENHANCEMENT)

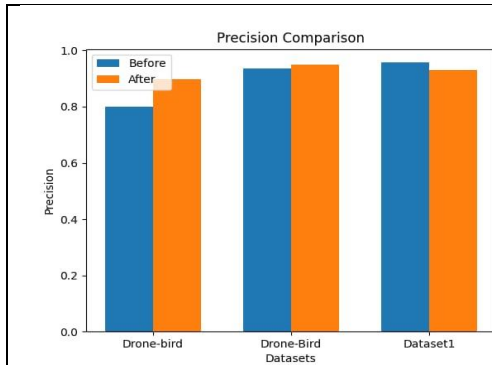


Fig 5.5.1 Precision comparison before and after applying dataset enhancement techniques, showing improved detection accuracy across datasets

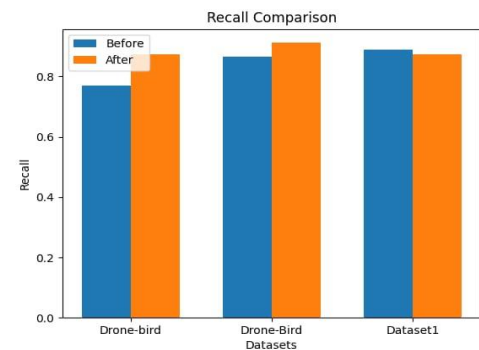


Fig 5.5.2 Recall comparison before and after dataset enhancement, indicating improved object detection sensitivity.

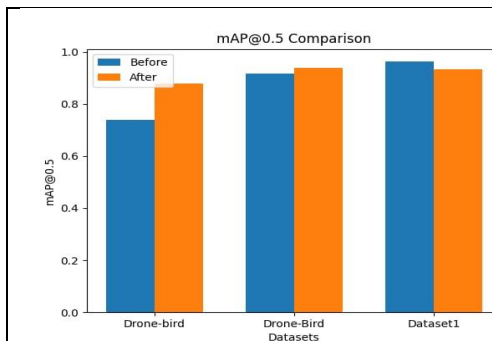


Fig 5.5.3 mAP@0.5 comparison before and after dataset enhancement, reflecting improved detection performance at IoU threshold 0.5.

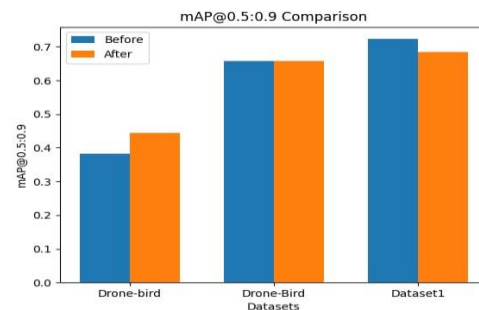


Fig 5.5.4 mAP@0.5:0.95 comparison before and after dataset enhancement, demonstrating improved model generalization across multiple IoU thresholds.

The side-by-side bar graphs present a comparative analysis of performance metrics before and after applying the Rolling Guidance Filter. It can be observed that most datasets show improvement in Precision and Recall after enhancement, indicating better detection accuracy and completeness. The $mAP@0.5:0.9$ graph further highlights improved localization performance in certain datasets. However, slight decreases in some cases suggest that enhancement may remove fine details in already high-quality images. Overall, the graphical comparison confirms that image enhancement improves model robustness and performance in challenging scenarios.

5.6 SUMMARY OF RESULTS AND DISCUSSIONS

The results and discussions presented in this chapter provide a comprehensive and systematic evaluation of the proposed drone detection system using the YOLOv5 model under various experimental conditions. The analysis was carried out in multiple stages, including model comparison, multi-dataset training, cross-dataset evaluation, and performance enhancement using image preprocessing techniques. Each stage contributed to a deeper understanding of the model's behavior, strengths, and limitations.

In the initial phase, different YOLO architectures were compared to identify the most suitable model for drone detection. The experimental results clearly indicated that YOLOv5 outperformed other models in terms of achieving a balanced trade-off between Precision, Recall, and Mean Average Precision (mAP). Its ability to maintain high detection accuracy while minimizing false positives and false negatives made it a reliable choice for further experimentation. This selection laid a strong foundation for subsequent analyses.

The evaluation of YOLOv5 on multiple drone datasets demonstrated its adaptability to varying data conditions. The model showed consistent performance across datasets with different sizes, image resolutions, and environmental complexities. Higher performance was observed in datasets with more uniform and less complex backgrounds, while slight variations occurred in datasets with diverse lighting conditions, occlusions, and background clutter. These observations highlight the sensitivity of object detection models to dataset characteristics and emphasize the importance of dataset quality and diversity.

To further assess the robustness of the model, cross-dataset evaluation was conducted. This experiment simulated real-world scenarios where a trained model is applied to previously unseen data. The results revealed that YOLOv5 maintained reasonable detection accuracy even when evaluated on new datasets, confirming its generalization capability. However, a slight reduction in performance was observed compared to training datasets, which can be attributed to domain shifts such as differences in camera angles, object scales, and environmental conditions. This behavior is expected in practical applications and reinforces the need for diverse training data.

In order to improve detection performance, an image enhancement technique known as the Rolling Guidance Filter (RGF) was applied to the datasets. This preprocessing step aimed to reduce noise and remove small-scale textures while preserving important structural details such as object boundaries. The enhanced datasets were used to retrain the YOLOv5 model, and the results were compared with those obtained from the original datasets.

The comparative analysis between original and enhanced datasets demonstrated that image enhancement plays a significant role in improving model performance, particularly in challenging visual conditions. Notable improvements were observed in Precision and mAP values for datasets with higher noise and background complexity, indicating that the model was able to better distinguish drone objects from irrelevant features. The graphical analysis, including Precision-Recall curves, Confidence curves, and F1-score curves, further supported these findings by showing smoother convergence and more stable performance after enhancement.

However, the results also indicated that image enhancement does not universally improve performance across all datasets. In cases where the original dataset already contained high-quality images with clear features, slight decreases in performance were observed after enhancement. This can be attributed to the potential loss of fine texture details during filtering. These findings highlight an important insight: preprocessing techniques should be applied selectively based on dataset characteristics rather than uniformly across all data.

Overall, the results obtained from both tabular and graphical analyses confirm that YOLOv5 is a robust and efficient model for drone detection tasks. Its ability to perform

well across different datasets, maintain generalization capability, and benefit from preprocessing techniques makes it suitable for real-world applications. Furthermore, the study demonstrates that combining deep learning models with appropriate image enhancement techniques can significantly improve detection accuracy and reliability.

This chapter provides a detailed analysis of the experimental results, which forms the basis for the overall conclusions presented in the next chapter.

CHAPTER 6

CONCLUSION

This project presents an enhanced real-time drone detection system using deep learning-based YOLO models, with a focus on improving detection accuracy and reducing misclassification between drones and birds. A detailed experimental study was conducted by training and evaluating multiple YOLO versions, including YOLOv5, YOLOv8, YOLOv10, and YOLOv11, using both individual and combined datasets.

From the comparative analysis in Chapter 5, it is evident that YOLOv5 outperformed other models in terms of overall stability and detection performance. While newer models showed competitive results, YOLOv5 consistently achieved a better balance between precision, recall, and mAP, making it more suitable for this specific application.

The cross-dataset evaluation results highlight a key limitation of models trained on single datasets, where performance dropped when tested on unseen data. This indicates poor generalization capability when dataset diversity is limited. However, after applying dataset enhancement techniques, including dataset merging, correction of label inconsistencies, and data augmentation, there was a significant improvement in model performance. The enhanced dataset resulted in:

- Higher precision, reducing false positives (incorrect detection of birds as drones)
- Improved recall, ensuring better detection of actual drone instances
- Increase in mAP@0.5 and mAP@0.5:0.95, demonstrating improved detection accuracy across different IoU thresholds
- Better model generalization across multiple datasets

The comparative results clearly demonstrate that dataset quality and diversity have a greater impact on model performance than the choice of model alone. The improvements observed after dataset enhancement confirm that proper annotation, class balance, and data diversity are critical factors in achieving reliable detection.

Additionally, the system showed improved capability in handling real-world challenges such as:

- Small object detection (distant drones)
- Complex backgrounds (sky, trees, buildings)
- Variations in lighting and environmental conditions.

The proposed system utilizes the powerful object detection capability of YOLO to identify drones accurately from images and video streams. One of the major advantages of YOLO is its ability to perform detection in a single stage, which allows the model to process images quickly while maintaining good accuracy. This makes it particularly suitable for real-time applications where immediate detection and response are necessary. The system is capable of analyzing continuous video frames and detecting drones by generating bounding boxes around the detected objects along with their confidence scores.

During the implementation of this project, the model was trained using drone image datasets that included different drone models, angles, and environmental conditions. Through this training process, the neural network learns important visual features such as drone structure, propellers, and movement patterns. As a result, the trained model becomes capable of distinguishing drones from other flying objects such as birds or airplanes. This ability to differentiate between similar objects significantly improves the reliability of the detection system.

Overall, the proposed system successfully achieves its objective of enhancing drone detection accuracy using deep learning. The results validate that YOLOv5, when trained on a well-structured and diverse dataset, provides robust, accurate, and efficient performance suitable for real-time surveillance applications.

The experimental results obtained from the project demonstrate that the YOLO-based detection system provides fast and accurate performance when compared to traditional computer vision techniques. Traditional approaches often rely on handcrafted features and multi-stage detection methods, which may result in slower processing times and reduced accuracy. In contrast, deep learning-based models automatically learn features from data, allowing them to adapt to different environments and complex backgrounds.

Another important achievement of this project is the ability to perform detection in real time. Real-time processing ensures that drone activities can be identified instantly,

allowing security personnel to respond quickly to potential threats. The system can also be integrated with existing surveillance infrastructure, making it a practical solution for real-world security applications.

In addition, the project emphasizes the growing role of machine learning models in solving real-world problems. The successful implementation of the YOLO-based detection framework proves that deep learning technologies can significantly enhance the accuracy, speed, and reliability of automated detection systems. As drone technology continues to evolve, intelligent detection systems such as the one proposed in this project will become increasingly important for maintaining safety and regulatory compliance.

Overall, the “Enhanced Real-Time Drone Detection Using YOLO” project successfully demonstrates how modern deep learning techniques can be applied to build efficient and reliable surveillance systems. The results indicate that YOLO-based object detection models provide a strong foundation for developing advanced drone monitoring solutions capable of operating in real-time environments.

The project not only contributes to the field of computer vision but also provides a practical solution for addressing the growing challenges associated with drone security. With continuous advancements in artificial intelligence, data availability, and computing power, such intelligent detection systems will play a crucial role in the future of automated surveillance and airspace monitoring.

CHAPTER 7

FUTURE SCOPE

Although the proposed system achieves promising results in real-time drone detection, there are several opportunities to further enhance its capabilities and expand its applications in the future.

One important area for improvement is the enhancement of detection accuracy. Future research can focus on collecting and training the model with larger and more diverse datasets that include drones of different shapes, sizes, and flight patterns under various environmental conditions such as fog, rain, low light, and long distances. This will help the system generalize better and reduce false detections.

Another important extension is the integration of object tracking algorithms with the detection system. Algorithms such as Deep SORT or SORT can be used to continuously track detected drones across multiple video frames. This will allow the system not only to detect drones but also to monitor their movement paths, speed, and trajectory over time.

The system can also benefit from using more advanced versions of the YOLO architecture such as YOLOv8 or YOLOv9. These models offer improved accuracy, better feature extraction, and enhanced performance compared to earlier versions, which can further improve the reliability of drone detection.

Another promising direction is the deployment of the system on edge computing devices. Platforms such as NVIDIA Jetson Nano or Raspberry Pi 5 can enable real-time drone detection directly at the surveillance location without the need for powerful centralized servers. This can make the system more portable, cost-effective, and suitable for field applications.

Future systems can also incorporate multi-sensor technologies to improve detection reliability. For example, combining visual detection with radar sensors, acoustic sensors, or thermal cameras can help detect drones even in conditions where normal cameras may struggle, such as during night time or poor weather conditions.

Another potential advancement is the development of automated alert and response systems. Once a drone is detected, the system could automatically send alerts to security personnel or activate countermeasures such as drone tracking, warning systems, or controlled airspace restrictions. This can significantly improve the safety of sensitive

locations like airports, government buildings, military bases, and public events.

In addition, future research can explore the use of artificial intelligence-based behaviour analysis to identify suspicious drone activities. By analyzing flight patterns and behaviour, the system may be able to distinguish between authorized and unauthorized drones, improving the overall security management process.

Finally, the system can be integrated with smart city surveillance networks and IoT platforms for large-scale monitoring and management of drone activities across urban areas.

In conclusion, the future scope of this project is vast, and continuous advancements in deep learning, computer vision, and edge computing technologies will further enhance the effectiveness and applications of real-time drone detection systems. monitoring, and smart city infrastructure. With further improvements in hardware acceleration and optimized neural network architectures, the system can achieve even higher performance levels.

In addition, the project emphasizes the growing role of machine learning models in solving real-world problems. The successful implementation of the YOLO-based detection framework proves that deep learning technologies can significantly enhance the accuracy, speed, and reliability of automated detection systems. As drone technology continues to evolve, intelligent detection systems such as the one proposed in this project will become increasingly important for maintaining safety and regulatory compliance.

Overall, the “Enhanced Real-Time Drone Detection Using YOLO” project successfully demonstrates how modern deep learning techniques can be applied to build efficient and reliable surveillance systems. The results indicate that YOLO-based object detection models provide a strong foundation for developing advanced drone monitoring solutions capable of operating in real-time environments. The project not only contributes to the field of computer vision but also provides a practical solution for addressing the growing challenges associated with drone security. With continuous advancements in artificial intelligence, data availability, and computing power, such intelligent detection systems will play a crucial role in the future of automated surveillance and airspace monitoring.

REFERENCES

- [1] Chen, C., Zheng, Z., Xu, T., Guo, S., Feng, S., Yao, W. and Lan, Y., 2023. Yolo-based uav technology: A review of the research and its applications. *Drones*, 7(3), p.190.
- [2] Chakraborty, D., Bej, S., Mukherjee, R., Agarwal, D. and Maity, S., 2023. Alokha Vision: A Deep learning based Automatic Drone Detection Approach for Cross Border Surveillance. *TAM METİN BİLDİRİ KİTABI*, p.184.
- [3] Chakraborty, D., Bej, S., Mukherjee, R., Agarwal, D. and Maity, S., 2023. Alokha Vision: A Deep learning based Automatic Drone Detection Approach for Cross Border Surveillance. *TAM METİN BİLDİRİ KİTABI*, p.184.
- [4] Zhang, Y., Rahman, F., Han, R., Dawes, J.M., Mildren, R., Nguyen, P. and Abir, T.A., The Laser-Based Drone Vision Disruption with a Real-Time Tracking System for Privacy-Preservation.
- [5] Ali, E.A. and Ilyas, M., 2022, October. Analyzing Optical Fibre Communication Networks for Intrusion Detection Using FBG Sensors. In *2022 International Symposium on Multidisciplinary Studies and Innovative Technologies (ISMSIT)* (pp. 788-793). IEEE.
- [6] Khazai, S. and Mirzaei, S., 2024. A Comparative Analysis of YOLO and Faster R-CNN Algorithms for Micro-UAVs Detection in Surveillance Videos.
- [7] Wang, Y., 2023, November. Attitude Control of Steel Structure Damage Detection under YOLO Algorithm and PID. In *2023 International Conference on Integrated Intelligence and Communication Systems (ICIICS)* (pp. 1-6). IEEE.
- [8] Webber, L., 2024. Real-Time UAS Detection in High-Security Environments.
- [9] Urda, D. and Herrero, Á., 2024, November. Human Detection in Images Captured by UAVs in Natural Environments. In *The 19th International Conference on Soft Computing Models in Industrial and Environmental Applications SOCO 2024: Salamanca, Spain, October 9-11, 2024 Proceedings, Volume 1* (Vol. 1, p. 3). Springer Nature.
- [10] Riadi, I., Yudhana, A. and Agustian, H., 2025, December. real-time vision-based drone detection for airspace security in restricted and critical infrastructure areas using deep learning. in *Serumpun International Conference Proceedings (SICP)* (Vol. 1, No. 2, pp. 173-182).
- [11] Yang, X. and Zhang, H., 2023, June. Foreign object debris detection based on improved YOLOv5 algorithm. In *International conference on image, signal processing*,

and pattern recognition (ISPP 2023) (Vol. 12707, pp. 1155-1160). SPIE.

[12] De Miguel, F., Caubilla, D., Cambra, C., Basurto, N., Urda, D. and Herrero, Á., 2024, October. Human Detection in Images Captured by UAVs in Natural Environments. In International Conference on Soft Computing Models in Industrial and Environmental Applications (pp. 3-12). Cham: Springer Nature Switzerland.

[13] Tai, N.T., Thang, B.D. and Hung, N.N., 2025. YOLOXpress: A lightweight real-time unmanned aerial vehicle detection algorithm. International Journal of Systematic Innovation, 9(2), p.85.

[14] Dash, S., Parida, P. and Kiran, V.K., 2025. An Improved Object Detection Algorithm for Maritime Search and Rescue Based on Drone Imagery. In AI and Blockchain in Smart Grids (pp. 249-259). Auerbach Publications.

[15] He, X., Guo, W. and Zhang, J., 2024, April. Task path planning system for autonomous feeding robots in three-dimensional farms underYOLOV3 algorithm. In 2024 Third International Conference on Distributed Computing and Electrical Circuits and Electronics (ICDCECE) (pp. 1-5). IEEE.

[16] Alsanad, H.R., Al_Hassani, R.T., Alrawi, A.A., Graovac, S. and Al Mashhadany, Y., 2025. Smart Surveillance of Waste Disposal Sites Using Image Processing and Artificial Intelligence System for Public Health Safety. Anbar Journal of Engineering Sciences, 16(2), pp.42-53.

[17] Manikandan, M., Shaji, N.S. and George, S.V., 2026. Real time drone surveillance system—for detection, distance estimation and automatic target lock using YOLO. In ITM Web of Conferences (Vol. 82, p. 03016). EDP Sciences.

[18] Li, M., Mao, S., Wang, H., Shi, Z. and Zhou, C., 2023, August. A training-time friendly network for real-time drone detection. In 2023 IEEE/CIC International Conference on Communications in China (ICCC) (pp. 1-6). IEEE.

[19] Liu, P., Zhang, Y., Zhang, K., Zhang, P. and Li, M., 2022, November. An improved YOLOv3 algorithm and intruder detection on transmission line. In 2022 China Automation Congress (CAC) (pp. 5736-5741). IEEE.

[20] Al-refai, G., Al-refai, M., El-moaqet, H., Ryalat, M. and Almtireen, N., 2025, May. Performance Evaluation of YOLOv7 for Object Detection in Dark Environments. In 2025 IEEE International Conference on Electro Information Technology (eIT) (pp. 036-041).

IEEE.

- [21] Ayuningtyas, A., Riadi, I. and Yudhana, A., 2025. The Effect of Light Intensity, Camera Pixel Quality, Camera Distance, and Object Altitude on Detection Accuracy in a Real-Time Drone Surveillance System Using YOLOv5. *Journal of Innovation Information Technology and Application (JINITA)*, 7(2), pp.234-246.
- [23] Hermanus, D.R., Supangkat, S.H. and Hidayat, F., 2025. Mini Drone-Based Precision Agriculture for Indonesian MSMEs: A Low-Cost AI-Assisted Monitoring System. *Jurnal Sistem Cerdas*, 8(2), pp.203-215.
- [24] Dong, X. and Lan, J., 2024. Optimal path for automated pedestrian detection: image deblurring algorithm based on generative adversarial network. *Journal of Measurements in Engineering*, 12(2), pp.298-311.
- [25] Meena, B.P. and Manikandan, M.S., 2025, June. Effectiveness of YOLO11-Based Lightweight Model for Drone Detection in Noisy Environments. In *2025 17th International Conference on Electronics, Computers and Artificial Intelligence (ECAI)* (pp. 1-8). IEEE
- [26] Zhai, X., Huang, Z., Li, T., Liu, H. and Wang, S., 2023. YOLO-Drone: an optimized YOLOv8 network for tiny UAV object detection. *Electronics*, 12(17), p.3664.
- [27] Zhang, Z., 2023. Drone-YOLO: An efficient neural network method for target detection in drone images. *Drones*, 7(8), p.526.
- [28] Kim, J.H., Kim, N. and Won, C.S., 2023, June. High-speed drone detection based on yolo-v8. In *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (pp. 1-2). IEEE.
- [29] Chen, C., Zheng, Z., Xu, T., Guo, S., Feng, S., Yao, W. and Lan, Y., 2023. Yolo-based uav technology: A review of the research and its applications. *Drones*, 7(3), p.190.
- [30] Chen, C., Zheng, Z., Xu, T., Guo, S., Feng, S., Yao, W. and Lan, Y., 2023. Yolo-based uav technology: A review of the research and its applications. *Drones*, 7(3), p.190.
- [31] Sahin, O. and Ozer, S., 2021, July. Yolodrone: Improved yolo architecture for object detection in drone images. In *2021 44th international conference on telecommunications and signal processing (TSP)* (pp. 361-365). IEEE.
- [32] Shandilya, S.K., Srivastav, A., Yemets, K., Datta, A. and Nagar, A.K., 2023. YOLO-based segmented dataset for drone vs. bird detection for deep and machine learning algorithms. *Data in Brief*, 50, p.109355.

- [33] Kucukayan, G. and Karacan, H., 2024. Yolo-ihd: Improved real-time human detection system for indoor drones. *Sensors*, 24(3), p.922.
- [34] Alkentar, S.M., Alsahwa, B., Assalem, A. and Karakolla, D., 2021. Practical comparison of the accuracy and speed of YOLO, SSD and Faster RCNN for drone detection. *Journal of Engineering*, 27(8), pp.19-31.
- [35] Patil, S., Jaybhaye, S.M., Khalifa, M.M., Kharche, S., Khatib, A. and Kshirsagar, A., 2023, December. Drone detection using YOLO. In *AIP Conference Proceedings* (Vol. 2938, No. 1, p. 020002). AIP Publishing LLC.
- [36] Madasamy, K., Shanmuganathan, V., Kandasamy, V., Lee, M.Y. and Thangadurai, M., 2021. OSDDY: embedded system-based object surveillance detection system with small drone using deep YOLO. *EURASIP Journal on Image and Video Processing*, 2021(1), p.19.
- [37] Caputo, S., Castellano, G., Greco, F., Mencar, C., Petti, N. and Vessio, G., 2021, December. Human detection in drone images using YOLO for search-and-rescue operations. In *International conference of the Italian association for artificial intelligence* (pp. 326-337). Cham: Springer International Publishing.